

DESIGNING EXPERIMENTS TO DISTINGUISH THEORIES*

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Abstract

We develop a measure of an experimental design's ability to distinguish between competing parametric theories. A design separates a theory from its competitors when these competitors are unable to reproduce behavior predicted by the theory. We propose an algorithmic procedure for constructing maximally-separating experimental designs, given a domain and hypothesized parameter values. Using this procedure, we generate 60 algorithmic designs to separate competing theories of choice under risk. We show experimentally that designs with high separation generate stronger statistical results; and that experimental results are more likely to favor a given theory when separation is asymmetric, i.e. high for the theory, low for its competitor. We then analyze separation in a novel dataset of 119 lottery choice experiments from the experimental risk literature. We provide evidence that algorithmic designs improve upon the state of the art for design development and demonstrate how design separation can change the interpretation of experimental results.

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1 Introduction

Empirical tests are a fundamental building block of theory. Good tests helps us understand *how* a theory fails, which guides innovation. Designing insightful ways to test theory has been a core contribution of experimental economics in particular. Typically, researchers rely on intuition or analytical insights to construct experimental tests. There are at least two problems with this. First, researchers may be leaving experimental power on the table if intuitive tests are far from the optimal tests. Second, intuitive tests may unfairly advantage some theories over others, which can lead researchers to misleading conclusions.

In this paper, we propose a measure of an experimental design's power to statistically test between theories. The intuition behind our measure is that we value testing theories on designs where competing theories make very different predictions. Suppose a researcher wants to test Expected Utility Theory (EUT) by asking subjects to make two choices between lotteries. She considers the two designs in Figure 1.

Design X		Design X'	
Task 1		Task 1	
Option A	Option B	Option A	Option B
\$28 w.p. 100%	\$0 w.p. 5% \$28 w.p. 75% \$40 w.p. 20%	\$30 w.p. 100%	\$0 w.p. 20% \$40 w.p. 80%
Task 2		Task 2	
Option A	Option B	Option A	Option B
\$0 w.p. 75% \$28 w.p. 25%	\$0 w.p. 80% \$40 w.p. 20%	\$30 w.p. 75% \$0 w.p. 5% \$40 w.p. 20%	\$0 w.p. 20% \$40 w.p. 80%

Figure 1: Choice tasks under Design X and Design X'.

For both designs, EUT predicts that subjects prefer the same option in both tasks. The Design X tasks are related by a common consequence, and the Design X' tasks are related by a common mixture. When we think about which design is a better test of EUT, we should ask: what behavior would we expect to see if preferences did *not* satisfy EUT? Concretely, we'd prefer a design where alternative theories of risk preference predict EUT-violating behavior. It turns out that on Design X, Cumulative Prospect Theory (CPT) (Tversky and Kahneman, 1992) predicts behavior which violates EUT; and on Design X', CPT predicts

behavior which is consistent with EUT.¹

If the researcher were to run Design X' , we might guess before seeing any data that behavior is likely to satisfy EUT; but this would tell us very little about whether EUT is a better description of behavior than CPT in general. Even if CPT were true, we would find that EUT describes behavior well on Design X' – but if we moved to Design X , EUT’s failure would be apparent. This is not merely a theoretical issue. The Design X' tasks appear in the experiment of Castillo et al. (2018), which is used to study subjects risk preferences. Based on experimental data, the paper concludes “expected utility is as good as other models at explaining behavior.”² Design X comes from Starmer and Sugden (1991), and in contrast, the experiment reveals that subjects are *not* consistent with EUT. These different conclusions need not be interpreted as fundamental inconsistencies in behavior; but instead as differences in what the experimental designs were set up to reveal.

Guided by this insight, we develop a formal criterion for experimental design which captures a design’s ability to statistically test between competing theories. A design is better according to this criterion if, supposing one theory is true, competing theories cannot explain the resulting data well. We informally refer to this criterion as *design power* to separate theories. We then propose an algorithmic procedure for experimental design, wherein the researcher inputs a set of theories and a space of candidate designs, and the algorithm selects a design within that space which best separates the theories.

We illustrate the usefulness of both the criterion and our design algorithm in a lottery choice experiment. In the experiment, we show that (i) designs with greater ex-ante ability to distinguish theories have stronger statistical results after we collect data; and (ii) when designs ex-ante advantage one theory over another, statistical results tend to favor the advantaged theory.

Finally, we use our framework to evaluate and analyze the existing experimental literature on lottery choice. We compute design power for a set of 119 designs from 84 papers. We observe suggestive evidence that our criterion is aligned with researchers’ design goals – but even so, algorithmic designs improve upon even the best designs from the literature. We look specifically at papers which test Expected Utility Theory (EUT) and make two observations. First, the papers which reach positive conclusions on EUT are relatively “low-powered” vis-a-vis CPT, much like Design X' in our earlier example. Second, we see that many low-powered designs actually *do* decisively reject EUT. This is interesting for a differ-

¹Predictions are evaluated for a range of parameter values estimated in the experimental risk literature.

²The set of other models considered includes Cumulative Prospect Theory. The experiment includes several additional tasks which we have excluded here for expositional ease, but across the full set it remains true that CPT-predicted behavior is EUT-consistent for most tasks.

ent reason. In such cases, EUT is failing; but CPT is not a good explanation for EUT’s failure. These insights are materially useful from the perspective of advancing theory, but hard to see without a formal language for assessing what a design is set up to test.

In Section 2, we define our design criterion and connect it to statistical power for model selection. Formally, we treat theories as parametric model classes. An individual model within a model class specifies a distribution of behavior for any potential experimental task. For instance, if the experimental domain is binary choice tasks between monetary lotteries, one relevant model class might be CRRA expected utility with a tremble. Fixing a risk aversion parameter and a tremble probability, each model within this class specifies computable choice probabilities on every possible lottery pair.

We define a design’s *normalized separation* between a fixed model and a single rival model class to be the normalized Kullback-Leibler (KL) divergence between the fixed model and the best-approximating model (in terms of KL divergence) from the rival class. For multiple models, the definition is similar in spirit. If the fixed model were true, then the normalized separation along with the number of subjects would exactly determine the experiment’s statistical power to select the correct model class over its rivals in a log likelihood horserace. We think of normalized separation as capturing the component of an experiment’s power which is attributable to the choice of design.

To calculate normalized separation between a fixed CPT model from EUT on a lottery choice design, we would (1) compute choice probabilities using our fixed CPT parameter values; (2) compute choice probabilities under the EUT parameter values that most closely match in terms of KL-divergence; and (3) calculate the variance-normalized KL divergence between these two distributions, which we divide. In many cases, researchers will be interested in separability across a range of fixed models – this could be different parameter values under a single model class, or different model classes altogether. In such cases, we take the relevant measure to be a weighted average under the fixed models of interest. Our proposed design criterion (“design power”) is a weighted average of normalized separation, with weights on plausible fixed models specified by the researcher.

We will sometimes be interested in separately analyzing separation and power depending on *which* model class we take as fixed. For two equally-weighted model classes M_1 and M_2 , it is possible that two different designs could have the same design power; but one’s power comes from separating M_1 from M_2 , and the other’s power comes from separating M_2 from M_1 . When we analyze this type of variation, we will use M_1 -design power and M_2 -design power to refer to average normalized separation computed only over the fixed models in the indicated class. We call a design “asymmetrically powered” if M_1 -design power

is high and M_2 -design power is low, or vice versa.

In Section 3, we treat experimental design as an optimization problem and implement an algorithm to find high-powered designs. The researcher specifies plausible models, competing model classes, and a design space; the goal is to find the design in this space that best separates the hypothesized model from the competing classes. In general, this is a challenging optimization problem for several reasons. First, our design criterion is non-convex, which means gradient-based methods will struggle to effectively traverse the landscape. Second, the design space is typically high-dimensional, which means exhaustive grid-based approaches are infeasible. Finally, the objective is costly to calculate, as each calculation requires an optimization sub-procedure to find the KL-minimizing parameters of the competing model classes. Making this calculation on a fixed design is easily manageable; but making the same calculation thousands of times throughout an optimization routine can be prohibitively slow.

In light of these challenges, we deploy Trust Region Bayesian Optimization (TuRBO) to solve our design optimization (Eriksson et al., 2019). Bayesian optimization is a standard approach for handling costly objective functions, in which rather than compute the full objective with each optimization step, the routine maintains a surrogate objective which can be cheaply calculated. The surrogate is used to identify promising points in the search space, the full objective is computed on sufficiently promising points, the surrogate function updates in response to the new information, and the search continues. TuRBO uses the same machinery but additionally deploys random restarts when a region repeatedly fails to show improvement. This feature helps manage the size and non-convexity of our search space. Because of the difficulty of our optimization problem, TuRBO cannot guarantee that we find the globally optimal point. In our view, this is not necessarily a problem; separation is important, but there’s nothing conceptually special about the maximally separating design.

In Section 4, we deploy our algorithmic design procedure to generate a set of 60 lottery choice designs, which we use to experimentally test two predictions of our theoretical framework. We consider 6 distinct theories of choice under risk and generate algorithmic designs optimized for different design criteria corresponding to different theory-pairs. Some designs are optimized to separate a larger model (e.g. CPT) from a smaller model class (e.g. EUT); some are optimized to mutually separate two non-nested model classes; and some are optimized to distinguish one model class from its competitor, without being able to separate in the reverse direction.

We use these 60 designs to empirically validate the consequences of separation for experimental results. First, we check that for 8 different theory-pairs, design power positively

predicts the strength of statistical results. For each design and for each theory-pair, we first calculate design power, and then use observed data to calculate the Vuong model selection z-statistic. We show that power positively predicts the absolute value of this test statistic for all 8 theory-pairs, with 4 pairs significant at $p = 0.05$. To give a sense of magnitudes, on our 8-task designs with 30 subjects, moving from a design with median CPT-EU power to one with maximum power increased the z-statistic for rejecting EU in favor of CPT by 1.6 units.

Second, we demonstrate how asymmetry in ex-ante separation generates spurious variation in directional results. For each design and for each of six non-nested theory-pairs, we calculate the difference between M_1 -design power and M_2 -design power. A positive difference indicates that the experiment is better-designed to separate M_1 from M_2 than to separate M_2 from M_1 ; a negative difference indicates the reverse. We again use Vuong statistic, where a positive value indicates more evidence for M_1 , and a negative value indicates more evidence for M_2 . Again, we find positive correlations for all 6 theory-pairs, with 5 pairs significant at $p = 0.05$. For some pairs, this effect is so extreme that moving from a design where M_2 has the greatest separability advantage to a design where M_1 has the greatest separability advantage can take the researcher from statistically rejecting M_1 in favor of M_1 to the reverse conclusion.

In Section 5, we take a step back to assess existing experiments from the risk literature in light of design power. We construct a novel dataset of 119 distinct lottery choice experimental designs from 84 different papers and assess the ability of these designs to distinguish between competing models of risk. We manually classify papers according to which theories (if any) the experiments purport to test, and we indeed find that experiments testing EU and CPT (or another form of rank-dependent utility) on average have greater normalized CPT-EUT design power than experiments which do *not* make such a claim. This suggests that our criterion is aligned with researchers' goals, and that researchers are able to recognize better designs over worse designs. However, our algorithmic designs far outperform even the *best* designs from the literature, both on CPT-EUT power and for the 7 other theory-pair criteria from our experiment. This indicates to us that there is meaningful scope for researchers to cheaply improve their experimental designs by deploying our algorithmic procedure.

Finally, we look at the subset of designs which test EUT and try to understand how considering design power leads us to new insights on how to interpret results. We document that all of the papers which came to a positive conclusions about EUT had relatively *poor* separation of CPT and EUT on their designs. This means that even if subjects exhibited CPT-like preferences, we wouldn't expect to see large deviations from EUT on these designs.

We then discuss the set of papers which document clear rejections of EUT, but do not have designs with high CPT-EUT separation. This indicates that although EUT fails on these designs, CPT is not a good description of how EUT fails. We analyze in particular the classical Allais paradox, which famously demonstrates a failure of EUT that is often attributed to CPT. We show that the Allais paradox is a low-power design according to our criterion. This reflects that at standard parameter values, CPT does not predict the typical pattern of EUT violations, an incongruence first pointed out by Neilson and Stowe (2002). In our view, this means that the failure of EUT on Allais paradox requires a different theoretical explanation. We think of this as an example of an *intuitive* test of EUT that can technically be rationalized under CPT; but it is not the test a researcher would design if they had CPT in mind.

Our framework for experimental design relates to several strands of prior literature. Our design criterion is conceptually consistent with Bronars Power (Bronars, 1987) and restrictiveness (Fudenberg et al., 2023), which are both measures of how well alternative distributions of behavior accord with theory on a given design. We take our measure one step further by turning it into an objective function for experimental design.

A number of papers in economics and cognitive science discuss objectives for experimental design and consider notions of efficiency and power specific to behavioral economics (Chapman et al., 2024; Snowberg and Yariv, 2025; Healy and Leo, 2024; Bland, 2025; Niederle, 2025; Baliatti et al., 2021; Myung and Pitt, 2009). We provide a concrete design criterion derived from statistical power for standard empirical exercises and propose a practical implementation to suit a general set of researcher goals.

Formally, our criterion is closely related to prior work on optimal design in econometrics and statistics. This literature proposed choosing designs on which alternative models poorly fit data generated by the hypothesized true model, first for regression models, and later for arbitrary distributions using KL divergence to pseudo-true alternatives (Atkinson and Fedorov, 1975; López-Fidalgo et al., 2007). Our unnormalized separation measure corresponds to a prior-averaged version of this KL criterion, following Bayesian design strategies proposed by Ponce De Leon and Atkinson (1991); Tommasi and López-Fidalgo (2010); Dette et al. (2015). To facilitate a clear power interpretation, we follow Vuong (1989) and normalize the KL divergence to an interpretable standard-normal scale. In addition, we clarify a number of practical questions that arise in model selection among behavioral models, including the role of parameter identification, alternative model-selection practices, and extensions to heterogeneity analysis, making the framework adaptable to the needs of applications in behavioral economics. We also derive empirically testable predictions linking

the theoretical design-power criterion to the behavior of model-selection evidence observed ex post.

Finally, we relate to a literature on how algorithms can be deployed to discover theoretically interesting experiments for theory testing. Both Fudenberg and Liang (2019) and Mulinathan and Rambachan (2023) propose using algorithms trained on existing behavioral data to generate new experiments on which known theories are predicted to fail. We view our framework as offering a complementary approach to guide design in settings where the researcher does not yet have rich data on how behavior varies with task features.

2 Theoretical Framework

We model a researcher choosing a set of experimental tasks to test between competing theories. The statistical power of an experiment depends both on its size and its component tasks. If two theories make nearly the same predictions on the experimental tasks, even a large dataset may not distinguish them. If the predictions are far apart, a small dataset can produce clear evidence. This section makes that idea precise, turns it into a practical design criterion, and derives empirical relationships which we test later on.

2.1 Model Separation and Statistical Power

We define an experimental design as a collection of T tasks. Let $\mathcal{X}$ be the task space and let $\mathbf{X} = (X_1, \dots, X_T) \in \mathcal{X}^T$ denote a design. For example, a task may be a binary choice between two lotteries, in which case an experiment is a collection of T lottery pairs. Let $\mathcal{Y}$ denote the behavioral response space associated with a task. For a binary choice experiment, $\mathcal{Y} = \{0, 1\}$. When an experiment is run, N subjects complete each task in $\mathbf{X}$ and produce $n = NT$ observations, which we denote $\mathbf{Y} = \{Y_{it} : i \leq N, t \leq T\}$. For a binary choice experiment, we observe $Y_{it} \in \{0, 1\}$ for each subject i and each task t .

We formalize “theories” as parametric model classes, and we model a researcher who will use experimental data to separate K competing classes. Formally, a model class M_k is a collection of models

$$M_k = \{g_k(y | x, \theta_k) : \theta_k \in \Theta_k\}, \quad \Theta_k \subseteq \mathbb{R}^{\ell_k},$$

where a model (M_k, θ_k) specifies a distribution of behavior $Y \sim g_k(\cdot | X, \theta_k)$ for any task $X \in \mathcal{X}$. Each model assumes behavior is independent across individuals i and tasks t . The

true model is fixed but is unknown to the researcher.

Example 1 (Risk Preferences in Lottery Choice).

Consider a researcher who wants to test between Cumulative Prospect Theory (CPT) and Expected Utility Theory (EUT) using binary choices between lotteries. Her task space $\mathcal{X}$ is choices of the form:

- Option A:** \$ w with probability p , \$0 otherwise
Option B: \$1 for sure

She formulates the theories CPT and EUT as parametric model classes, where parameters pin down binary choice probabilities as a function of $X = (w, p)$. We demonstrate using Prelec probability weighting function for CPT (Prelec, 1998), and a power utility function and fixed tremble probability for both. Probability of choosing A under each model is given by:

$$g_{CPT}(A | X, \theta_{CPT}) = \begin{cases} 1 - \epsilon, & \exp(-(-\ln p)^\gamma) \cdot w^\alpha \geq 1 \\ \epsilon, & \text{otherwise} \end{cases}$$

$$g_{EUT}(A | X, \theta_{EUT}) = \begin{cases} 1 - \tau, & p \cdot w^\alpha \geq 1 \\ \tau, & \text{otherwise} \end{cases}$$

where $\tau \in [0, \frac{1}{2})$ is the probability of a trembling-hand error. ◇

A standard approach to model comparison is to fit each class by maximum likelihood and compare the fitted log likelihoods.³ With two classes, 1 and 2, a positive log-likelihood ratio favors M_1 and a negative ratio favors M_2 . A Vuong test uses the same log-likelihood ratio after dividing by its estimated sampling variation. This puts the comparison on a standard-normal scale, so its magnitude can be read against familiar critical values (Vuong, 1989).

We derive our design criterion from the expression for an experiment's power to correctly select model class M_k over its competitors using these approaches, in a world where the true model is (M_k, θ_k) . To reach this expression, we first define pseudo-true models and model separation.

³To fix ideas, we present this section using likelihood-based model selection. However, our ideas extend to generic statistical decision theory and other data analysis procedures, as discussed in Appendix A.1. For example, GARP experiments have traditionally used model selection rules counting strict GARP violations, which are not likelihood-based but can still be accommodated within generic decision theory.

Definition 1 (Pseudo-true distribution).

Assuming the data on each design task X_t is generated by some distribution $p(\cdot | X_t)$, a pseudo-true parameter of class j is any

$$\theta_j^*(p, \mathbf{X}) \in \arg \min_{\theta \in \Theta_j} \frac{1}{T} \sum_{t=1}^T \text{KL}(p(\cdot | X_t) \| g_j(\cdot | X_t, \theta)),$$

where $\text{KL}(f \| g) = \int f(y) \log[f(y)/g(y)] dy$. The model $(M_j, \theta_j^*(p, \mathbf{X}))$ is the pseudo-true model of p , which we refer to as the pseudo-true distribution.

The pseudo-true model for p from model class M_j is the model that minimizes Kullback–Leibler divergence.⁴ For $p = g_k(\cdot | \cdot, \theta_k)$, we abuse notation slightly by writing $\theta_j^*(\theta_k, \mathbf{X})$ to denote the pseudo-true parameter of M_j . The model $g_j(\cdot | X_t, \theta_j^*(\theta_k, \mathbf{X}))$ thus denotes M_j 's best approximation of model (M_k, θ_k) on the tasks in design $\mathbf{X}$.

We next define the *separation* between a fixed model and a rival model class, which will be the basis of our design criterion.

Definition 2 (Separation and normalized separation).

Fix a model (M_k, θ_k) , a rival class M_j , and a design $\mathbf{X}$. For a single observation Y generated by the model on task $X_t \in \mathbf{X}$, define the log-likelihood ratio of (M_k, θ_k) and the pseudo-true model from M_j :

$$L_t^{kj} = \log \frac{g_k(Y | X_t, \theta_k)}{g_j(Y | X_t, \theta_j^*(\theta_k, \mathbf{X}))}, \quad Y \sim g_k(\cdot | X_t, \theta_k). \quad (1)$$

The separation of (M_k, θ_k) from M_j on $\mathbf{X}$ is

$$\bar{D}_{kj}(\theta_k; \mathbf{X}) = \frac{1}{T} \sum_{t=1}^T \mathbb{E}[L_t^{kj}].$$

The normalized separation is

$$\bar{\rho}_{kj}(\theta_k; \mathbf{X}) = \frac{\bar{D}_{kj}(\theta_k; \mathbf{X})}{\sqrt{\bar{V}_{kj}(\theta_k; \mathbf{X})}}, \quad \bar{V}_{kj}(\theta_k; \mathbf{X}) = \frac{1}{T} \sum_{t=1}^T \text{Var}(L_t^{kj}). \quad (2)$$

For $K > 2$, let $\bar{\rho}_k \equiv (\bar{\rho}_{kj})_{j \neq k}$ and define $(\bar{R}_k)_{jj'} = \frac{1}{T} \sum_t \text{Cov}(L_t^{kj}, L_t^{kj'}) / \sqrt{\bar{V}_{kj} \bar{V}_{kj'}}$.

$\mathbb{E}[L_t^{kj}]$ is the KL-divergence of the fixed model from the pseudo-true rival model on X_t , which makes separation $\bar{D}_{kj}$ the task-level average KL-divergence. When we compare

⁴Note that the pseudo-true model is not in general unique, but for later results we will require all pseudo-true models to induce the same distribution of behavior on fixed tasks $X_t \in \mathbf{X}$.

theories by assessing the fit of parametric models, it's not the fit of an arbitrary model in M_j that matters – it's the fit of the *best* model in M_j . If the empirical distribution matched (M_k, θ_k) , this would be $(M_j, \theta_j^*(\theta_k, \mathbf{X}))$. Separation is a measure of how much evidence an average observation from design $\mathbf{X}$ contributes to separating these two models.

The normalization in $\bar{\rho}_{kj}$ accounts for how variable that evidence is. Two designs can have the same average KL distance even though one produces steady evidence and the other relies on rare, large likelihood differences. Dividing by $\sqrt{\bar{V}_{kj}}$ expresses separation in units of its own sampling noise and therefore distinguishes the two. When $K > 2$, $\bar{R}_k$ captures how the evidence against different rivals co-moves because all pairwise comparisons use the same outcomes.

Example 2 (Risk Preferences in Lottery Choice, Continued).

Fix a CPT model with $\theta_{CPT} = (\gamma_0, \alpha_0, \tau_0)$. Define q_t to be the probability of choosing A on X_t under (M_{CPT}, θ_{CPT}) , and q_t^* to be the probability of choosing A on X_t under the pseudo-true model from M_{EUT} :

$$\begin{aligned} q_t &= g_{CPT}(A | X_t, \theta_{CPT}) \\ q_t^* &= g_{EUT}(A | X_t, \theta_{EUT}^*(\theta_{CPT}, \mathbf{X})) \end{aligned}$$

The *separation* of (M_{CPT}, θ_{CPT}) from M_{EUT} on $\mathbf{X}$ is

$$\bar{D}(\theta_{CPT}; \mathbf{X}) = \frac{1}{T} \sum_{t=1}^T q_t \log \frac{q_t}{q_t^*} + (1 - q_t) \log \frac{1 - q_t}{1 - q_t^*}. \quad (3)$$

The *normalized separation* is

$$\bar{\rho}(\theta_{CPT}; \mathbf{X}) = \frac{\bar{D}(\theta_{CPT}; \mathbf{X})}{\sqrt{\bar{V}(\theta_{CPT}; \mathbf{X})}}, \quad \bar{V}(\theta_{CPT}; \mathbf{X}) = \frac{1}{T} \sum_{t=1}^T q_t(1 - q_t) \left[\log \frac{q_t(1 - q_t^*)}{q_t^*(1 - q_t)} \right]^2 \quad (4)$$

◇

Normalized separation is our design criterion of choice because of its formal connection to an experiment's statistical power for model selection. The following proposition makes this connection explicit. Given the data, let $\hat{\ell}_j$ be model class j 's maximized average log likelihood, let $\hat{k}$ be the class with the largest fitted likelihood, and, fixing class k as the truth, define $\hat{\mathbf{D}}_k = (\hat{\ell}_k - \hat{\ell}_j)_{j \neq k}$ and $\bar{\mathbf{D}}_k = (\bar{D}_{kj})_{j \neq k}$ as vectors of empirical and theoretical separations. For a vector v , write $\Phi_R(v) = \Pr(W \leq v)$ for $W \sim \mathcal{N}(0, R)$, with the inequality applied coordinate by coordinate.

Proposition 1 (Power of a design).

Fix a model (M_k, θ_k) and a design $\mathbf{X}$ with T tasks. Suppose the following conditions hold:

- (a) For every alternate model class M_j , every pseudo-true parameter θ_j^* induces the same distribution on the tasks in $\mathbf{X}$.
- (b) For every $j \neq k$, $0 < \bar{V}_{kj} < \infty$.
- (c) Every model class satisfies the likelihood regularity conditions stated in Appendix I.1, ensuring first-order equivalence between its maximized likelihood and the likelihood at its pseudo-true distribution.

Then, as $N \rightarrow \infty$,

$$\sqrt{NT} (\hat{\mathbf{D}}_k - \bar{\mathbf{D}}_k(\theta_k; \mathbf{X})) \xrightarrow{d} \mathcal{N}(0, \bar{\Sigma}_k),$$

where

$$(\bar{\Sigma}_k)_{jj'} = \frac{1}{T} \sum_{t=1}^T \text{Cov}(L_t^{kj}, L_t^{kj'}), \quad (\bar{\Sigma}_k)_{jj} = \bar{V}_{kj}.$$

Consequently,

$$\Pr(\hat{k} = k \mid X, (k, \theta_k)) \approx \Phi_{\bar{R}_k}(\sqrt{NT} \bar{\rho}_k(\theta_k; \mathbf{X})). \quad (5)$$

For two classes, this is the ordinary univariate normal probability $\Phi(\sqrt{NT} \bar{\rho}_k)$.

Condition (a) allows the pseudo-true parameter to be non-unique, so long as all such parameters make the same predictions on the tasks in $\mathbf{X}$. Condition (b) requires each likelihood comparison to have finite, nonzero sampling variation, and in particular, no rival model can reproduce the generating distribution exactly on the design. These are standard regularity and separation conditions for likelihood comparisons, stated at the fixed scenario rather than imposed globally on the model classes.

Proposition 1 has a direct frequentist interpretation. If the true model were known to be (M_k, θ_k) , then the experiment's statistical power to correctly select M_k over all $M_{j \neq k}$ as N goes to infinity is given by Equation 5. Power is increasing in N , the number of subjects, and in $\bar{\rho}_k$, the normalized separation. We think of normalized separation as the component of an experiment's power attributable to the choice of design. Although Proposition 1 is stated for maximum likelihood selection, the same first-order power expression applies when the researcher uses a penalized selection method such as AIC and BIC for fixed-dimensional models, provided that the penalty is on a scale of $o(\sqrt{NT})$ relative to the total log-likelihoods.

Importantly, for any fixed model and rival model classes, $\bar{\rho}(\theta_k; \mathbf{X})$ can be calculated *before seeing any data*. Under the hypothesis that data is generated under (M_k, θ_k) , a design

with higher normalized separation leads to greater experimental power.

This is the same logic as a textbook power calculation. For a one-sided test based on $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2/n)$,

$$\Pr(\text{reject } H_0 \mid \beta) \approx \Phi\left(\sqrt{n} \frac{\beta}{\sigma} - z_\alpha\right). \quad (6)$$

The normalized treatment effect β/σ determines how quickly power rises with sample size. Here $\bar{\rho}_k$ plays the same role: statistically decisive conclusions are easier when the design makes the relevant normalized separation large.

2.2 Choosing a Design Ex Ante

If the researcher knows the true model, the design objective is clear: she should simply choose the tasks that maximize normalized separation from rival classes. The practical difficulty is that the true model is not known, and may not even belong to the researcher’s consideration set—this is why the experiment is needed. This is true in standard power calculations as well: in order to calculate power, a researcher must specify the effect size they want to be powered for, often called a “minimal detectable effect.” In our case, the researcher will specify the models they want to be able to separate from one another. To simplify notation, the remainder of this section considers the case of selecting between two models ($K = 2$), denoted $\mathcal{M}_1$ and $\mathcal{M}_2$.⁵

Our proposed design criterion requires the researcher to supply a hierarchical prior over parameters within a model class, and over the model classes themselves. First, for both model classes M_k , the researcher supplies a parameter grid $(\theta_{ks})_{s=1}^S$ with nonnegative weights (w_{ks}) that sum to 1.⁶ The directional average normalized separations are

$$\bar{\rho}_1(\mathbf{X}) = \sum_{s=1}^{S_1} w_{1s} \bar{\rho}_{12}(\theta_{1s}; \mathbf{X}), \quad \bar{\rho}_2(\mathbf{X}) = \sum_{s=1}^{S_2} w_{2s} \bar{\rho}_{21}(\theta_{2s}; \mathbf{X}).$$

Thus $\bar{\rho}_1(\mathbf{X})$ measures normalized separation when Theory 1 generates the data, while $\bar{\rho}_2(\mathbf{X})$ does the same for Theory 2. Second, the researcher chooses model weights $P_1, P_2 \geq 0$, with $P_1 + P_2 = 1$. These are weights chosen in advance by the researcher, not estimates formed

⁵For general K , see Appendix C.

⁶We use a discrete grid for computational reasons, though conceptually there’s no reason a researcher shouldn’t use a continuous prior over Θ_k .

from the new data. The overall design criterion is

$$\mathbf{X}^* \in \arg \max_{\mathbf{X} \in \mathcal{X}_\ell^T} \bar{\rho}(\mathbf{X}), \quad \bar{\rho}(\mathbf{X}) = P_1 \bar{\rho}_1(\mathbf{X}) + P_2 \bar{\rho}_2(\mathbf{X}), \quad (7)$$

where $\mathcal{X}_\ell^T$ is the feasible design space. The recipe is therefore: specify the two models, their parameter grids and weights, and the feasible tasks; compute the best-fitting rival and $\bar{\rho}$ for every candidate scenario; average as in (7); and search for the maximizing design.

Our criterion is closely related to prior work on optimal design for model selection. For a fixed scenario (M_k, θ_k) , $\bar{D}_{kj}(\theta_k; \mathbf{X})$ is the KL discrimination criterion studied by López-Fidalgo et al. (2007) and averaging $\bar{D}_{kj}$ over generating parameters yields the Bayesian criteria studied by Ponce De Leon and Atkinson (1991); Tommasi and López-Fidalgo (2010); Dette et al. (2015). Our objective $\bar{\rho}(\mathbf{X})$ instead averages over the normalized likelihood-ratio following Vuong (1989), and therefore puts KL separation on the standard-normal scale that governs model-selection power.

One could think of the hierarchical prior as encoding the researcher's beliefs about the true data generating process. These may be guided, for instance, by estimates from prior literature, or economic calibration assumptions. However, our preferred interpretation is that the hierarchical prior simply encodes which models the researcher would like to be powered for. For instance, some researchers may only care about power to reject M_2 if M_1 is true, in which case they would set $(P_1, P_2) = (1, 0)$. Others may only care about separating the models in certain regions of the parameter space. Below, we discuss several example scenarios and discuss how we would formulate the design objective in our framework.

Example 3 (Risk Preferences in Lottery Choice, Continued).

To formulate a design criterion, the researcher specifies her hierarchical prior over the EUT and CPT model classes. First, she sets priors over the classes. For nested classes like CPT and EUT, it is natural to set $P_{CPT} = 1$ and $P_{EUT} = 0$. The large class can mechanically fit any data generated by a model in the small class; so the scenario in which we hope to separate the models is when the truth is outside the small class. Intuitively, we want a design where EUT is unable to fit the data well if there is truly probability distortion as in CPT.

Next, she chooses CPT parameters based on which models within the CPT class she wants to be able to separate from EUT. The probability weighting parameter γ in CPT is typically estimated between 0.4 and 1, where $\gamma = 1$ gets back to EUT. The power utility parameter α is typically estimated between 0.6 and 1, with 1 being risk neutral. Suppose she wants to be able to separate CPT from EUT under moderate or extreme probability weighting, with moderate or no risk aversion and moderate trembling. Then, she could use

γ	α	τ
0.6	1	0.25
0.6	0.7	0.25
0.8	1	0.25
0.8	0.7	0.25

Table 1: Grid of θ_{CPT} values to be used in design objective.

the following grid, applying equal weights to each row: She does not need to specify a parameter grid for EUT, since P_{EUT} is equal to 0.

For this researcher, $\bar{\rho}(\mathbf{X}) = \bar{\rho}_{CPT}(\mathbf{X})$ is the average normalized separation, defined in Equation 4 of the four θ_{CPT} values in Table 1. $\diamond$

Example 4 (Restrictiveness on Initial Play in Matrix Games).

Consider a researcher studying initial play in 3x3 matrix games. She wants to assess the Logit Level-1 model from Fudenberg et al. (2023) et al. in an experiment, and she wants her set of experimental games to restrict the model’s flexibility as much as possible. Let M_1 be the model class of *eligible* behaviors, defined as all distributions over the 3 actions such that strictly dominant moves are played with at least 1/3 probability and strictly dominated moves played with at most 1/3 probability. Let M_2 be the Logit Level-1 model. In this case, M_1 encodes an agnostic prior over models which the researcher would like to test against. We propose setting $P_1 = 1$, $P_2 = 0$, and placing equal weight on each distribution in M_1 . We use the the resulting average normalized separation of M_1 from rival class M_2 as the design criterion.

Note that in this setting, separation in our framework is closely analogous to the restrictiveness criterion proposed by Fudenberg et al. (2023). Our preferred criterion is prior-averaged normalized separation $\bar{\rho}$ due to its connection to standard inference critical values, but we observe that the average unnormalized separation $\bar{D}$ coincides with the numerator of the restrictiveness criterion proposed by Fudenberg et al. (2023) up to orientation of the KL divergence. In practice, the two criteria often induce a similar order over designs. The primary difference is how we think about what these objects represent: Fudenberg et al. (2023) emphasizes restrictiveness as a property of theory, while we emphasize (normalized) separation as a property of a design. $\diamond$

2.3 From Ex Ante Power to Ex Post Evidence

The design calculation uses stylized models and hypothesized parameters. Its practical value therefore depends on whether ex ante separation predicts the evidence that real data

produce. We first derive that link under the maintained model and then test whether it survives approximation and misspecification.

Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be the maximum-likelihood estimates for the two models and define

$$\hat{D} = \frac{1}{NT} \sum_{i,t} \log \frac{g_1(Y_{it} | X_t, \hat{\theta}_1)}{g_2(Y_{it} | X_t, \hat{\theta}_2)}, \quad \hat{z} = \frac{\sqrt{NT} \hat{D}}{\hat{\sigma}}. \quad (8)$$

Here $\hat{\sigma}^2$ is the sample variance of log-likelihood ratios. $\hat{D}$ is the mean log-likelihood ratio, and $\hat{z}$ is its Vuong standardization. Both are common model-selection statistics in empirical work.

For the mean log-likelihood ratio, define the weighted directional quantities

$$\mathbb{D}_1(\mathbf{X}) = \sum_s w_{1s} \bar{D}_{12}(\theta_{1s}; \mathbf{X}), \quad \mathbb{D}_2(\mathbf{X}) = \sum_s w_{2s} \bar{D}_{21}(\theta_{2s}; \mathbf{X}).$$

Corollary 1 (Ex ante separation predicts ex post evidence).

Under the conditions of Proposition 1:

(i) Conditional on truth $(1, \theta_1)$,

$$\hat{D} = \bar{D}_{12}(\theta_1) + O_p(n^{-1/2}), \quad \hat{z} = \sqrt{n} \bar{\rho}_{12}(\theta_1) + O_p(1).$$

Under truth $(2, \theta_2)$, the centers are instead $-\bar{D}_{21}(\theta_2)$ and $-\sqrt{n} \bar{\rho}_{21}(\theta_2)$.

(ii) If the model and scenario weights form a hierarchical prior and the supported scenarios have enough moments for expectations to converge, then

$$\begin{aligned} \mathbb{E}_w[\hat{D} | X] &\longrightarrow P_1 \mathbb{D}_1 - P_2 \mathbb{D}_2, & \mathbb{E}_w[|\hat{D}| | X] &\longrightarrow P_1 \mathbb{D}_1 + P_2 \mathbb{D}_2, \\ \frac{\mathbb{E}_w[\hat{z} | X]}{\sqrt{n}} &\longrightarrow P_1 \bar{\rho}_1 - P_2 \bar{\rho}_2, & \frac{\mathbb{E}_w[|\hat{z}| | X]}{\sqrt{n}} &\longrightarrow P_1 \bar{\rho}_1 + P_2 \bar{\rho}_2. \end{aligned}$$

Part (i) states that, for a fixed truth, each ex post statistic is centered on its matching ex ante quantity design by design. Part (ii) gives the corresponding result after averaging over scenarios. These statements suggest that, when the true model is known or the scenarios of models and parameters are exactly correctly specified, we should obtain a slope of one when we regress the ex post statistic on the ex ante power measure. Of course, we do not expect these conditions to hold exactly in practice, hence our empirical benchmark is therefore a *positive* slope, not a slope of one.

More importantly, whether the slope is positive answers the question of whether by choosing powerful designs ex ante, we predictably produce stronger statistical evidence.

This is the right test for assessing whether our ex ante power measure is informative for choosing designs, rather than a test of whether the parametric models, parameter grid, or scenario weights correctly describe the population.

Both sign and magnitude matter. The sign of $\hat{D}$ identifies the better-fitting model; $|\hat{D}|$ measures the strength of its advantage. The Vuong statistic has the same sign and expresses magnitude in estimated sampling-noise units. Thus directional separation predicts which model wins, while symmetric separation predicts how decisive the comparison is.

We evaluate these relationships using paired model comparisons on 60 experimental designs. Let d index designs and let n_d be the number of observations in design d . With equal model weights, we use the average directional separation to explain the strength of evidence and their difference to explain its direction:

$$|\hat{D}_d| = \alpha_D^{\text{str}} + \beta_D^{\text{str}} \frac{\mathbb{D}_{1,d} + \mathbb{D}_{2,d}}{2} + \varepsilon_{D,d}, \quad (9)$$

$$\hat{D}_d = \alpha_D^{\text{dir}} + \beta_D^{\text{dir}} \frac{\mathbb{D}_{1,d} - \mathbb{D}_{2,d}}{2} + \eta_{D,d}, \quad (10)$$

$$|\hat{z}_d| = \alpha_z^{\text{str}} + \beta_z^{\text{str}} \frac{\sqrt{n_d}(\bar{\rho}_{1,d} + \bar{\rho}_{2,d})}{2} + \varepsilon_{z,d}, \quad (11)$$

$$\hat{z}_d = \alpha_z^{\text{dir}} + \beta_z^{\text{dir}} \frac{\sqrt{n_d}(\bar{\rho}_{1,d} - \bar{\rho}_{2,d})}{2} + \eta_{z,d}. \quad (12)$$

In the empirical section, we take these predictions to the data using a lottery-choice experiment. Following the above specifications, we test whether designs with greater average separation produce stronger model-selection evidence and whether differences in directional separation predict which model the evidence favors.

2.4 Discussion

Parameter ranges in overlapping and nested model classes. In general, we allow model classes to overlap or be nested. However, we argue that the parameters that conceptually make sense for power calculations lie outside the overlapping region, since within the overlapping region the two model classes induce identical distributions of behavior and are indistinguishable on any design.⁷ As a result, we emphasize that that Proposition 1 is pointwise in the generating model and parameter. Condition (b), which requires $\bar{V}_{kj} > 0$, exactly corresponds to restricting our attention to parameters outside the overlapping region, where

⁷For example, in the case of EU versus CPT, a researcher trying to design a well-powered experiment necessarily cares about probability-weighting parameter γ away from 1, and the relevant notion of separation is the one-directional $\bar{\rho}_{\text{CPT}}$, as discussed in Example 3.

normalized separation and its power interpretation are well defined.

A separate subtlety regarding overlapping models lies in the downstream empirical analysis. Since maximum-likelihood model selection and Vuong tests are illustrative of common model-selection exercises, we use them as leading motivation for our power criterion in this section. However, both procedures require care in interpretation when model classes are nested or overlapping.⁸ Since our preferred objective is defined at the normalized-separation $\bar{\rho}$ level, these subtleties in downstream inference are not inherent to our framework. Researchers could instead use alternatives such as penalized likelihood selection or modified model-selection statistics (Shi, 2015; ?), and our ex ante power measure remains well motivated for these alternative downstream inference procedures.

Normalized separation as design criterion. In Section 2.2, we recommended to optimize over designs based on average normalized separation $\bar{\rho}(\mathbf{X})$. One could instead maximize weighted average power at a chosen sample size,

$$\text{WAP}(X; n) = P_1 \sum_s w_{1s} \Phi(\sqrt{n} \bar{\rho}_{12}(\theta_{1s}; \mathbf{X})) + P_2 \sum_s w_{2s} \Phi(\sqrt{n} \bar{\rho}_{21}(\theta_{2s}; \mathbf{X})). \quad (13)$$

We recommend (7) because it does not require N , and hence $n = NT$, to be settled early in the design process. Furthermore, note that if we worked with (13), we would place little value on additional separation when experimental power is already high, since the normal CDF flattens as its value approaches 1. In practice, additional separation is likely still useful. The following result states when the two criteria give the same ranking.

Lemma 1 (Ranking designs away from the power ceiling).

Let $\bar{\rho}(X_A) > \bar{\rho}(X_B) > 0$, fix $\beta \in (1/2, 1)$, and let $z_\beta = \Phi^{-1}(\beta)$. If

$$1 - \frac{\bar{\rho}(X_B)}{\bar{\rho}(X_A)} \geq \frac{z_\beta^2}{6},$$

then $\text{WAP}(X_A; n) > \text{WAP}(X_B; n)$ at every n for which every scenario-specific power under X_A is below β .

Average normalized separation and average power need not rank every pair of designs identically, however, Lemma 1 isolates the comparisons that matter most for improvement:

⁸It is well known that unpenalized maximum likelihood mechanically weakly favors the containing class and that the usual normal approximation for the classical Vuong statistic is not uniformly valid near the overlap of model classes.

when the designs’ power has not saturated and the gain in effect size is large, the rankings agree. Once some scenarios are already near certain selection, the ranking can depend on the chosen n because extra separation barely changes their power. In the following sections, we will abuse terminology slightly and refer to designs with high average normalized separation as highly powered designs for expositional simplicity.

Heterogeneity. Throughout this section, we discussed model selection under the assumption that subjects are homogeneous in their behaviors. This, of course, does not hold true. Researchers are often interested in analyzing heterogeneity across subjects and thinking of model selection more broadly as an exercise in studying the population shares of agents that are best described by different models. Appendix B provides a set of formal results and detailed discussion about common statistical procedures used to analyze heterogeneity and how our notion of design power extends to those settings.

The main takeaway is that design power typically matters for the statistical properties such as bias and variance of population share estimators. Importantly, using low-powered designs may induce asymptotic bias in these estimators that cannot be eliminated simply by increasing the number of subjects.

Model misspecification. For the purpose of choosing well-powered designs, Proposition 1 provides a literal power interpretation under the assumption that one of the specified models generates the data. This result provides an analytical benchmark, but in practice it is likely that *all* of the researcher’s models are misspecified. In this case, suppose $p_{0t} = p_0(\cdot \mid X_t)$ denotes the actual distribution of behavior on task X_t , and let $g_{jt}^* = g_j(\cdot \mid X_t, \theta_j^*(p_0, \mathbf{X}))$ denote class j ’s pseudo-true distribution on that task. Under analogous likelihood regularity conditions, we have $\hat{D} \xrightarrow{P} \Delta_0(\mathbf{X}) \equiv \frac{1}{T} \sum_{t=1}^T \{\text{KL}(p_{0t} \parallel g_{2t}^*) - \text{KL}(p_{0t} \parallel g_{1t}^*)\}$, so likelihood selection favors the class that is closer in average KL divergence to actual behavior on the tasks in $\mathbf{X}$ (Vuong, 1989). Importantly, note that the task-level true distributions p_{0t} and the KL gaps can change with $\mathbf{X}$. Under arbitrary misspecification, the best-approximating model class among the parametric models can therefore change across designs, even for a fixed population of subjects. Consequently, conclusions about whether one model class is better than another should be defined relative to the specific design that produced the empirical evidence, and we caution against interpreting model-comparison results as “global” rankings that extend beyond that design.

2.5 Practical Applications

We have motivated our design criterion in terms of statistical power for standard empirical exercises, but we believe the framework is practically useful for many stages of the scientific process. Before any experiments are run, searching for highly-powered designs can help theorists understand the empirical content of the considered model classes. At the time of experimentation, the criterion guides researchers on how to select designs. Finally, when evaluating experimental results, the criterion can offer insight how those results should be interpreted.

A high-powered design is a set of tasks which highlight distinctive behavioral features of the theories being considered. Given a tool to automatically search for such designs, we can use resulting high-powered designs to clarify where arbitrary parametric models diverge. In Section 3, we give an example of a high-powered design which concisely illustrates the core difference between the two considered theories. If the tool is unable to produce high-powered designs, this is informative as well – it tells us that on our search domain, the models are essentially the same.

As we have discussed, having a design criterion at the time of design can both help the researcher find designs which deliver decisive empirical results; and also help avoid selective sampling. Setting up the criterion forces the researcher to explicitly clarify the scope of domain under consideration as well as the models she hopes to disambiguate.

Not only should the design criterion inform where researchers run experiments, it should affect the interpretation of results. If an experiment shows that two theories fit the data similarly well, but the design is not powered to separate those two theories, then we are in fact learning very little. This is exactly the result we’d expect to find in the event that one of these theories is true, and the other generally fits poorly. Conversely, if experimental results reveal a clear winner on a low-powered design, this tells us that the priors used to calculate the design criterion are not consistent with the actual parameters needed to explain the data. This should lead us either to update our priors about which parameters are reasonable; or question whether the “winner” is really the right explanation of the observed behavior.

3 Algorithmic Design

The design objective in Equation (7) defines a black-box optimization problem over the feasible design space $\mathcal{X}^T$. Solving this optimization problem presents several practical challenges. In this section, we discuss some of these challenges and describe our preferred

approach, which utilizes Trust-region Bayesian Optimization (Eriksson et al., 2019). We present an example of a lottery choice design which comes out of our search algorithm.

3.1 Design Search with Trust-Region Bayesian Optimization

Several features of our design problem pose challenges for standard optimization methods. First, the objective typically has a complicated global geometry. Fixing a design and a candidate model, the value of the objective is given by the normalized separation between the candidate model and its best-fitting alternatives, and since the parameters and possibly the class of the best fitting alternative change with the design, the resulting objective value as a function of design can be non-concave, strongly anisotropic, and non-smooth. Hence standard gradient-based procedures are often sensitive to initialization values and converge to low quality local solutions.

Second, computing the objective is itself expensive. Given a design, we need to first calculate behavioral predictions under all maintained models; and then we need to run an optimization sub-routine to find the best approximating model(s) from the rival class(es). Computing this once for a fixed design is usually easy, but hundreds of iterative computations may be infeasible.

Finally, the design spaces is often high-dimensional. Consider a simple binary choice task between two lotteries, each with two states. This single task may have as many as six dimensions: two payoffs and one probability for each lottery. Even a 10-task design like this would have sixty dimensions – and this would be relatively small by experimental standards. The large size of the search space makes grid-based approaches infeasible.

Given these challenges, we adopt a Bayesian-optimization approach tailored to high-dimensional, multimodal objectives: trust-region Bayesian optimization (TuRBO, Eriksson et al., 2019.) Below, we describe the procedure at a high level. See Appendix E for precise mathematical details.

Bayesian optimization (BO) is a standard tool for optimization problems with costly objectives. Rather than optimize directly on the costly objective, BO works by probabilistically modeling the true objective. This probabilistic “surrogate model” is initialized with some prior over the objective at every point in the search space. An acquisition rule is deployed to select a promising design $\mathbf{X}$ for evaluation. The acquisition rule favors designs which, under the surrogate model, have either high mean, i.e. are expected to be good on average; or high variance, which may produce high draws. The costly objective is evaluated on $\mathbf{X}$, the surrogate model is updated in a Bayesian way using the learned value, and the process

repeats.

A known difficulty of conventional global BO in high-dimensional, multimodal problems is that a single surrogate may fail to approximate the entire space well, inducing large posterior uncertainty that push towards over-exploration rather than actually improving the objective value. Inspired by TuRBO (Eriksson et al., 2019), we restrict candidate search to adaptive local trust regions. A “trust region” is a hyperrectangle in the design space centered at the best design found within it, and our procedure maintains one or more trust regions in the search process. At each iteration, we fit a Gaussian-process surrogate to the evaluations made in the local round and selects the next evaluation points by Thompson sampling from the Gaussian-process posterior over the trust region. If several consecutive evaluations improve upon the region’s best evaluation, the trust region expands; if several consecutive proposals fail to improve, it contracts. When a region shrinks below a minimum side length, it is terminated and reinitialized elsewhere (e.g., seeded near a new promising point), which helps avoid spending the full budget refining an unpromising basin.

Our implementation of design search deploys TuRBO with a single trust region. For full details on our surrogate model and acquisition rule, see Appendix E. In computational experiments, this produced better results more quickly than alternatives methods (see Appendix Figure 8).

3.2 Discussion

While TuRBO is our preferred approach, we make several caveats. First, we note that because of the non-convexity and generality of our optimization problem, we cannot guarantee global optimality of the returned solution. We think such a guarantee is not essential to our goal of using algorithmic search to produce designs with high power. As long as the search substantially improves power relative to designs obtained otherwise, we view it as a helpful tool.

Second, a researcher’s particular objective may render other optimization approaches more useful. For instance, Mullainathan and Rambachan (2023) pose an optimization problem for generating experimental “anomalies” which has the same structure as ours, and thus many of the same practical issues. However, they are specifically interested in *minimal designs*, e.g. simple pairs of choice problems which illustrate a theory failure. In their implementation on pairs of lottery choices, they further reduce the dimension of the search space by fixing a subset of features and successfully optimize using a combination of grid-based and gradient-based approaches. We recommend TuRBO because it is robust to many com-

mon issues that we anticipate; but practitioners may find other optimization algorithms to be better suited to their specific environment.

3.3 Example Design

For the remainder of the paper, we turn to the domain of choice under risk to deploy our design framework. In Figure 2, we present a simple, two-task design optimized to separate Cumulative Prospect Theory from Expected Utility Theory.⁹ The task space $\mathcal{X}$ is binary choices between two-outcome lotteries in a common state space.

Choice 1		
Probability	Option A	Option B
0.85	\$99	\$74
0.15	\$0	\$99

Choice 2		
Probability	Option A	Option B
0.09	\$99	\$0
0.91	\$0	\$4

Figure 2: Two-task algorithmic design, highly-powered to distinguish CPT from EUT

To see why this design is highly-powered for this purpose, first look at Choice 1. Option A is a high chance of \$99, and a low chance of 0. Option B is a high chance of \$75, and a low chance of \$99. Option A has a higher expected value, but under CPT, subjects will overweight the state which pays \$0 with 15% and prefer Option B. EUT can accommodate this choice, but it requires the decision-maker to be moderately risk averse.

In Choice 2, Option A is a low chance of \$99 with \$0 otherwise, and Option B is a high chance of \$4 with \$0 otherwise. Now, CPT predicts that the high chance of \$0 in Option A is deflated, and the low chance of \$0 in Option B is inflated. This means that CPT predicts Option A is preferred to Option B. Notice that in the previous choice, EUT could accommodate the CPT prediction by increasing risk aversion; but in this choice, the CPT prediction is the riskier option. EUT can make sense of either choice individually; but can't make sense of the joint preference pattern predicted by CPT.

⁹The design objective uses logistic noise specifications of both models and is described explicitly in Section 4.

4 Experimental Design

To demonstrate the importance of design power when selecting experimental tasks, we turn to perhaps the most widely studied experimental paradigm in behavioral economics: binary choice between lotteries. The purpose of our experiment is to validate the relationships between ex ante power and ex post evidence discussed in Section 2.3. First, we want to show that designs with higher average normalized separation $\bar{\rho}$ generate stronger statistical results for model selection. This tells us that researchers who care about having highly-powered experiments materially benefit from applying our design criterion. Second, we want to show that the direction of statistical results, i.e. which model gets selected, depends in part on the relative power of the competing theories. When a design is highly-powered to separate Model 1 from Model 2, but low separation of Model 2 from Model 1, we tend to see evidence that is more favorable to Model 1. When the relative separation reverses, the statistical evidence shifts positively towards Model 2.

4.1 Design Construction

Since we are interested in how statistical results vary with power *across* designs, we construct 60 unique lottery choice designs optimized for 20 distinct objectives. A “design” in our experiment is a set of 8 choice tasks between two money lotteries. We decided on 8-task designs because we wanted designs large enough to identify model parameters; but small enough for an analyst to interpret which features of designs cause models to fail. In each task, both lotteries are defined in a common state space with three possible states. State probabilities are restricted to multiples of 0.05, and both lotteries’ payoffs are restricted to positive whole dollars between 0 and 50. The common state structure is explained to subjects and communicated visually in the experimental implementation. When we construct algorithmic designs, we search over sets of 8 binary choice tasks which meet these criteria.¹⁰

4.1.1 Power Objectives

Each of our 20 design power criteria involves 2 competing models of stochastic choice under risk. We consider a set of 6 models in total, described in brief below. Each model, when parameterized, specifies a binary choice probability for any lottery task. Full parametric specifications for these models can be found in Appendix G.1.

¹⁰We also add a requirement that no more than 2 tasks involve first-order stochastic dominance, which gives us more reliable identification in numerical checks.

1. **Expected Utility with Logit Noise (EU).** We assume a power utility function, i.e. $u(x) = x^\alpha$.
2. **Cumulative Prospect Theory with Logit Noise (CPT).** We again assume power utility, and we add a Prelec 1-parameter probability weighting function.
3. **Random Expected Utility with a Tremble (REU).** We assume a population of power utility parameters α which follow a lognormal distribution, and a fixed tremble probability.
4. **Random Cumulative Prospect Theory with a Tremble (CPT).** We again assume a lognormal distribution of power utility parameters with a fixed tremble probability, and we add a (non-random) Prelec 1-parameter probability weighting function.
5. **Comparison Complexity (CC).** We use the expected utility CDF-complexity model from Shubatt and Yang (2024), again assuming power utility.
6. **Salience with Logit Noise (SAL).** We use the continuous salience value function proposed by Bordalo et al. (2012).

Each of the objectives used to generate our designs is associated with a pair of theories (model classes) from this list. For two model classes M_1 and M_2 , recall that $\bar{\rho}_1$ is a design's power to select M_1 over M_2 if M_1 is the truth; and $\bar{\rho}_2$ is the design's power to select M_2 over M_1 if M_2 is the truth. Both of these objects depend on which parameter values we hypothesize as the "truth," which we discuss more below. There are three types of objectives we use, which we call *one-sided* power, *symmetric* power, and *asymmetric* power. We use *one-sided* power when M_1 and M_2 have a nesting relationship, such as with CPT and EU. Intuitively, if behavior is truly EU, CPT will be able to fit the data well on *every* design. From a theory-testing standpoint, the right notion of design power here is one-sided: we just want $\bar{\rho}_{CPT}$ to be as large as possible on our design, meaning EU will do a poor job trying to fit data from CPT agents. For M_1, M_2 without a nesting relationship, we usually care about *symmetric* design power, meaning we want both $\bar{\rho}_1$ and $\bar{\rho}_2$ to be as large as possible. We implement this by maximizing $\min\{\bar{\rho}_1, \bar{\rho}_2\}$.¹¹ For this experiment, we are also interested in designs optimized for *asymmetric* power on non-nested pairs. We would like to see how results vary across designs with large $\bar{\rho}_1$ and small $\bar{\rho}_2$, versus designs with large $\bar{\rho}_2$ and small $\bar{\rho}_1$. We implement this by maximizing either $\bar{\rho}_1 - \bar{\rho}_2$, or $\bar{\rho}_2 - \bar{\rho}_1$.

¹¹Another researcher may reasonably prefer an alternative approach to combining $\bar{\rho}_1$ and $\bar{\rho}_2$, such as a sum. We found that using a minimum produced more balanced designs, which we prefer for our purposes.

The theory pairs and associated design objectives are laid out in Table 2 below. By convention, we always label the alphabetically-first theory as Theory 1. In total, we have 20 distinct objectives, and we generate 3 different designs using each.

Theory 1	Theory 2	One-Sided $\bar{\rho}_1$	Symmetric $\min\{\bar{\rho}_1, \bar{\rho}_2\}$	Asymmetric 1 $\bar{\rho}_1 - \bar{\rho}_2$	Asymmetric 2 $\bar{\rho}_2 - \bar{\rho}_1$
CPT	EU	✓	-	-	-
RCPT	REU	✓	-	-	-
CC	CPT	-	✓	✓	✓
CC	RCPT	-	✓	✓	✓
CC	SAL	-	✓	✓	✓
CPT	RCPT	-	✓	✓	✓
CPT	SAL	-	✓	✓	✓
RCPT	SAL	-	✓	✓	✓

Table 2: Power objectives maximized in construction of lottery choice designs

4.1.2 Hypothesized Parameters

To calculate the value of these objectives for any given 8-task design, we need to specify parameter values. This is, in one sense, a large degree of freedom; a researcher may hypothesize any parameter values they like. But if the researcher is completely miscalibrated, the proof will be in the pudding: highly powered designs will *not* do a good job of distinguishing between competing theories. At the same time, hypothesized parameters need not be exactly right for these constructs to be useful. Recall that the researcher’s goal is to find designs which reveal clear winners; so if highly-powered designs do this, it’s an indication that the calibration is good enough to be valuable.

For each of our models, we hypothesized several sets of parameters θ guided by the previous literature. Some parameters have been estimated in many previous papers, such as the CRRA power utility parameter α , or the Prelec probability-weighting parameter γ . For these, we chose some benchmark values from the literature. For others, such as the continuous salience parameter δ , we could not find existing parametric estimates; here, we conducted calibration exercises on the paper’s experimental tests and used values consistent with these examples. To calibrate logit noise parameters, we fixed different preference parameters and used existing lottery choice datasets to come up with an appropriate level of noise. Since we hypothesize multiple parameter combinations for each model, we calcu-

late $\bar{\rho}_i$ by taking a simple average over design power for each row in the parameter grid.¹² The grid for each model was preregistered¹³ and can be found in Appendix G.1.

4.2 Experimental Procedures

We implemented our lottery choice tasks in a Qualtrics survey and recruited subjects through Prolific. We collect choice data on 480 distinct tasks (8 tasks from 60 different designs). After reading instructions and passing comprehension checks, our subjects encounter a random subset of 6 designs, or 48 tasks in total. They were not aware of the “design” groupings, but were simply told that they would encounter 48 choice tasks. Of the 521 subjects who passed our comprehension checks, 308 completed the entire survey and passed two attention checks to be included in our final sample. The number of subjects per task ranged from 26 to 38, with a median value of 31.

We explained the tasks to subjects by telling them that they had a choice between two lotteries, A and B, which both pay out real money depending on the color of a randomly drawn ball (red, blue, or green). We told them there would always be 20 balls in total, and that each task screen would inform them of the ball colors for that task and the payout amounts for both options. See Figure 3 for an example of a task display. Subjects were paid a \$4.50 completion fee and informed that they would have a one-in-ten chance at being selected for a bonus payment if they successfully passed all attention checks. If selected for a bonus, one of their 48 choices would be selected at random, their chosen lottery would be played out by the computer, and they would receive the monetary payout. Full instructions can be found in Appendix G.2. The median bonus payment was \$39 and the median completion time was 22 minutes.

4.3 Results

Our goal is to understand how statistical results vary across different designs as a function of a design’s power to separate theories. We focus on two statistics in particular: the log likelihood ratio (LLR) the Vuong z -statistic for model selection, which we referred to as $\hat{D}$ and $\hat{z}$ respectively in Section 2.3. To compute $\hat{D}$ and $\hat{z}$ on a given design for model classes M_1 , M_2 , we use maximum likelihood to estimate $\hat{\theta}_1$ and $\hat{\theta}_2$ on the data for that design and then follow Equation 8. For both statistics, a positive value indicates that Theory 1 fits

¹²Once again, another researcher may prefer a different aggregation mode, such as a minimum. We found that an average worked the best for our context.

¹³Preregistration available at <https://osf.io/xyab3/overview>

Probability		Option A	Option B
	50%	\$46	\$50
	40%	\$47	\$50
	10%	\$48	\$0

Figure 3: Example lottery choice task from our experiment.

the data better; a negative value indicates that Theory 2 fits the data better; and a larger absolute value indicates that the winning theory has a larger advantage over the losing theory.

4.3.1 Power and Strength of Results

Our first empirical goal is to test whether more powerful designs lead to stronger statistical evidence. In the first two columns of Table 3, we do this separately for each of our eight theory pairs. Specifically, for each pair, we calculate ex ante $\bar{\rho}$ and $\bar{D}$ on all 60 designs. For nested theories, we take $\bar{D}$ and $\bar{\rho}$ to be the one-sided objectives corresponding to the larger theory. For non-nested theory pairs, we weight both theories equally, i.e.

$$\begin{aligned}\bar{D} &= 0.5\bar{D}_1 + 0.5\bar{D}_2 \\ \bar{\rho} &= 0.5\bar{\rho}_1 + 0.5\bar{\rho}_2.\end{aligned}$$

Then, we compute the empirical statistics $\hat{D}$ and $\hat{z}$ for that pair using the observed data. Reported coefficients correspond to β_D^{str} and β_z^{str} from Equations 9 and 11 respectively.

We see that both coefficients are positive for every single theory pair, i.e. more highly powered designs are generating stronger statistical results. The strength of the relationship varies across different theory pairs, with 4 of 8 being significant at $p = 0.05$ for both specifications. To give a sense of magnitudes, the coefficient on $\bar{\rho}_{CPT}$ in row 1 implies that moving from the median design to the best design in terms of $\bar{\rho}$ increases the value of the model

Theory pair	Strength		Direction	
	$ \hat{D} \sim \bar{D}$	$ \hat{z} \sim \sqrt{NT} \bar{\rho}$	$\hat{D} \sim \bar{D}_\Delta$	$\hat{z} \sim \sqrt{NT} \bar{\rho}_\Delta$
CPT vs. EU	1.08*** (0.23)	0.69*** (0.23)		
RCPT vs. REU	0.40*** (0.07)	0.48*** (0.09)		
CC vs. CPT	0.74*** (0.10)	0.49*** (0.11)	1.44*** (0.16)	1.25*** (0.20)
CC vs. RCPT	0.34** (0.14)	0.27 (0.17)	0.55*** (0.18)	0.86*** (0.24)
CC vs. SAL	0.80*** (0.12)	0.68*** (0.14)	1.21*** (0.14)	1.36*** (0.16)
CPT vs. RCPT	0.33* (0.17)	0.08 (0.19)	1.24*** (0.38)	1.01** (0.43)
CPT vs. SAL	0.07 (0.16)	0.44* (0.26)	0.06 (0.21)	0.56 (0.36)
RCPT vs. SAL	0.37* (0.22)	0.08 (0.25)	0.96*** (0.35)	1.19** (0.46)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Table 3: Columns (1) and (2) report regression coefficients (s.e.) from regressing the empirical absolute log likelihood ratio and absolute Vuong statistic on separation $\bar{D}$ and normalized separation $\bar{\rho}$ respectively. Columns (3) and (4) report regression coefficients from regressing the signed statistics on the *difference* in separation between the two theories, as in Equations 10 and 12. Note that N is held roughly fixed across designs, so virtually all variation in the columns (2) and (4) come from $\bar{\rho}$.

selection t -statistic from 1.38 to 2.98.

4.3.2 Power and Direction of Results

Our second empirical goal is to show that asymmetric design power can tip the scales in favor of a theory, before any data is ever collected. Consider two non-nested models M_1 and M_2 and two designs X_1 and X_2 , where X_1 is asymmetrically powered in favor of M_1 (i.e. large $\bar{\rho}_1$, small $\bar{\rho}_2$) and X_2 is asymmetrically powered in favor of M_2 . This means that on X_1 , M_1 would be able to approximately fit data generated in truth from M_2 ; but M_2 would not be able to fit data generated from M_1 . The theories are not nested in general, but on this design, M_1 approximately nests M_2 . For X_2 , it's the reverse.

These power differences may generate differences in relative model fits for reasons entirely unrelated to the models' global fits. If one theory, say M_1 , is correctly specified, we expect to see that M_1 decisively beats M_2 on X_1 ; but both theories fit similarly well on X_2 . Sign reversals can then arise only from sampling noise and should not be large or persistent on designs that sharply separate the two model classes. If *both* theories are misspecified, however, the best-approximating class can change with the tasks, as discussed in Section 2.4, so we may well see M_1 win on X_1 , and M_2 win on X_2 . Crucially, such wins tell us very little about which model is better in general, or even which model features are more important on the design – instead, they reflect the relative flexibility of the models to explain data on the design.

To understand the empirical importance of such effects, we turn to our directional regressions using the *difference* in power within our six non-nested theory pairs, reported in Columns (3) and (4) of Table 3. We use the same outcome variables as the previous exercise, except now we use signed values. Our regressors are now *differences* in power across the two theories:

$$\begin{aligned}\bar{D}_\Delta &= \bar{D}_1 - \bar{D}_2 \\ \bar{\rho}_\Delta &= \bar{\rho}_1 - \bar{\rho}_2\end{aligned}$$

Positive coefficients in these regressions indicate that as ex-ante power tips in favor of Theory 1, ex-post statistical evidence tips in favor of Theory 1 as well. We see that the coefficients are again positive for all theory pairs, and significant at $p = 0.05$ for 5 of 6.¹⁴ This

¹⁴Note that we do not include the nested pairs for this exercise, since there would be no way for the smaller theory to be asymmetrically powered against the larger theory under our definitions.

effect is so large that we actually observe reversals in statistical conclusions as we vary $\bar{\rho}_\Delta$ for some theory pairs. For instance, in the three designs we constructed to asymmetrically power CC over RCPT, we observe test statistics 3.74, 3.80, and 2.31. For the three designs constructed to RCPT over CC, the test statistics are -2.31, -1.00, and -1.56. In our view, these differences reflect systematic variation across these designs in the relative constraints these two theories place on data.

5 Risk Literature

In Section 4, we demonstrated that design power has consequences for the statistical outcomes. Now, we want to provide evidence that algorithmic designs offer meaningful improvements over what researchers are already doing; and show how considering the power of existing designs can help us generate new insights as analysts. To do this, we assemble a first-of-its-kind dataset of 119 experimental designs in choice under risk, drawn from 84 unique papers and 38 years of research.

5.1 Dataset Assembly

There are many different experimental paradigms used to study risky choice behavior, but we narrow our focus to binary choice designs where subjects make at least two separate choices between a pair of lotteries. This restriction is due in part to feasibility constraints; but also limits us to designs in the same domain as our own experiments, which will allow us to make apples-to-apples design power comparisons. This means we exclude valuation and choice list designs, even if technically a choice list is implemented through a table of binary choices.¹⁵ We exclude ambiguity designs where probabilities or payoffs are not specified, and we also exclude designs where payoffs cannot be clearly ordered (e.g., some lotteries pay out apples and others pay out oranges).

We begin by compiling a list of candidate articles using the EconLit database. We pulled all journal articles satisfying one of the two following conditions:

1. JEL code d81 and one of codes {d91, c90, c91}. Code d81 is for “Criteria for Decision-Making under Risk and Uncertainty,” d91 is for “Micro-Based Behavioral Economics,” and c91 and c90 are both related to experiments.

¹⁵We consider choice lists to be a separate form of behavioral elicitation because binary choices in sequential price list rows are not plausibly independent, as is assumed in our theory.

2. Abstract contains a term related to risky choice and a term related to experiments. Risk terms: {gamble, risk preference, risk preferences, choice under risk, expected utility, lottery}. Experiment terms: {experiment, experiments, experimental, experimentally}.¹⁶

We limit ourselves to articles published in one of the Top 100 Economics journals, according to the IDEAS/RePEc aggregate rankings in December of 2025. This search and filter procedure gives us 1,217 candidate articles. We download PDFs for all of these articles and extract the first 8 pages of text. We use a large language model to scan this text and flag articles which discuss an experiment with binary lottery choices. 233 articles are flagged, and we go through each of these by hand to extract the experiment. In total, 84 articles include an extractable experimental design that meets our criteria.¹⁷

For each design, we record the probabilities and payoffs of both lotteries for each binary choice task. If the lotteries are communicated to participants in a common state space and the correlation structure is described in the text, this is recorded. Any compound structure is also recorded, though none of the theories we study here make explicit use of compoundness. If a paper contains distinct experiments which are analyzed separately, these are recorded as separate designs. If an experiment includes some tasks that meet our criteria and others that do not (e.g., some binary menus and some trinary menus), we include only the tasks that meet the criteria and record it as partial design. We do not record the payoff units, only numerical quantities as they are presented to participants.

We classify designs into 3 categories based on the nature of the empirical exercise:

1. *Comparison*, in which two or more models are explicitly compared based on the experimental results. This includes horseraces, mixture estimations, classification, or violation counts.
2. *Other Test*, in which at least one model is being meaningfully tested, but models are not explicitly being compared. This includes, for example, many experiments which test the independence axiom or the betweenness axiom.
3. *Measurement*, in which the purpose of the experiment is not to test a model, but to measure some kind of preferences. This includes experiments used to measure risk

¹⁶The reason we did not use “risk” on its own as a risk term is that it increased search results by an order of magnitude, and by inspection the additional articles seemed largely irrelevant to our interests.

¹⁷The most common reasons for excluding papers were ambiguous gambles as in Ellsberg experiments, re-analyses rather than new experiments, dynamically defined choices as in staircase elicitations, and theory-only papers. There were also several experiments which met our criteria, but in which the set of lottery tasks was not explicitly given in the paper or appendix.

aversion or skewness preferences.

Of the 119 designs, we classify 14 as *Comparison*, 82 as *Other Test*, and 23 as *Measurement*. We also record for each design whether Expected Utility and/or a form of Rank-Dependent Utility (such as Cumulative Prospect Theory) are discussed as being tested or inspiring the tests. We classified 65 theory-testing papers as testing Expected Utility, and 26 as testing a form of Rank-Dependent Utility.¹⁸

5.2 Computing Power

We can compute $\bar{\rho}$ on all of these designs for any theories we like. Because we are interested in how our algorithmic designs compare to the literature at large, we focus specifically on EU and CPT. If our algorithmic designs meaningfully improve upon the literature in terms of $\bar{\rho}_{CPT}$, which aligns with the explicit empirical goals in many of these papers, we take this as strong support for an algorithmic approach to design more generally.

In order to calculate and summarize $\bar{\rho}$, we must make a consequential choice on how to handle the fact that the scale of payoffs varies widely across experimental designs – a classic Allais paradox problem has a potential \$5,000,000 payout, while other experiments may have a max payout of \$0.50. Taken literally, any stochastic theory with Fechnerian noise (as in our formulation of CPT and EU) would predict very different choice probabilities on these two designs. Any noise level which generates meaningful variation in choice probabilities on the \$0.50 design will imply virtually deterministic choice on the \$5,000,000 design. However, no reasonable analyst would actually expect a parameterized model to extrapolate in this way. The noise is not the substantive part of these models; the preference parameters are. Since none of our theories are substantively about how choice changes with scale, we opt to standardize the payoff-level of all designs before calculating power. Specifically, for each design, we first divide all payoffs by $M/50$, where M is the maximum absolute payoff in the whole design. This puts each literature design on exactly the same payoff scale as our own experiments.

Another source of variation across these designs is the number of tasks. Intuitively, more tasks in a design should always increase expected model separation, since it generates strictly more data.¹⁹ However, recall that our unit of analysis $\bar{\rho}$ refers to the *average* power

¹⁸We do not include in this classification experiments designed to test only a specific axiom of Expected Utility, such as the betweenness axiom.

¹⁹Strictly speaking, more tasks need not increase the experiment’s power to correctly select the true model, if new tasks introduce sufficiently high variance.

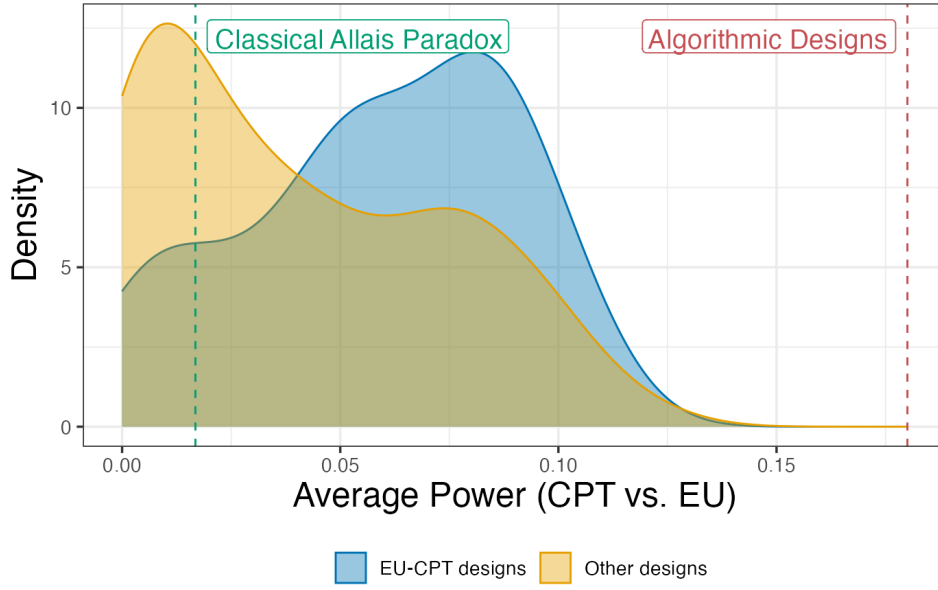


Figure 4: Average Power to Separate CPT from EU

contributed by each task in the design. Practically, $\bar{\rho}$ gives us an even playing field to compare designs of different sizes; but there are also two reasons why we might actually care more about this object than total power. The first reason is efficiency – if a design with many tasks has small $\bar{\rho}$, then even if there are sufficiently many tasks to separate theories, a researcher could save resources by running a smaller design with larger $\bar{\rho}$. The second reason has to do with *why* we run experimental tests of theory at all. A good experiment should illustrate clearly to an analyst how a theory fails and why. In our view, it is easier to draw insights from an experiment where a theory is far from the truth on average, as opposed to a large collection of small failures.

5.3 Analyzing Power

We begin by summarizing design power to separate CPT from Expected Utility. In Figure 4, we plot the literature’s distribution of power for separating CPT from EU, splitting designs into subgroups based on whether they explicitly test EU and CPT.

This figure illustrates several facts. First, designs explicitly focused on EU and CPT indeed have higher power to separate CPT from EU (mean $\bar{\rho}$ of 0.059 vs. 0.040, $p = 0.01$). This indicates to us that researchers in this literature are generally aligned with our proposed objective, and have a real sense of which design features make experiments more powerful. Second, we see that our algorithmic designs ($\bar{\rho} = 0.18$) far outpace even the best

literature designs in terms of power ($\bar{\rho} = 0.12$). In our experiment from Section 4, a change in $\bar{\rho}$ of this size for CPT-EU was associated with an increase of 0.65 in the z-statistic for selecting CPT over EU. This suggests that the current approach to design is leaving power on the table, which our algorithmic procedure could recoup.

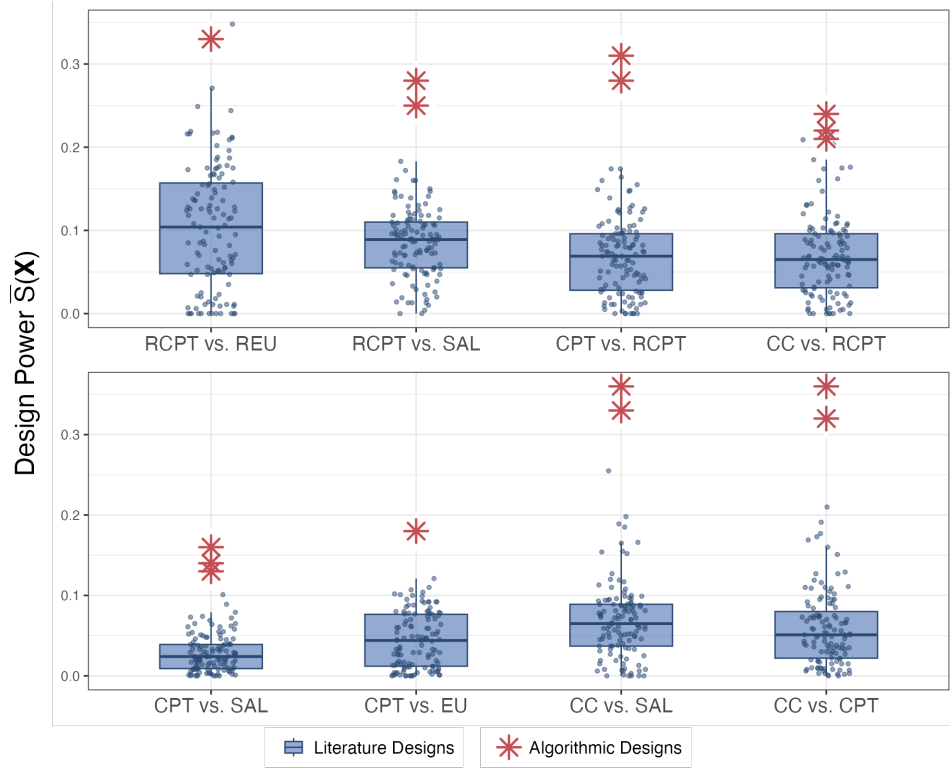
It is impossible to say definitively that existing designs are suboptimal given the researcher’s goals, which we cannot observe. Many papers are not intended to separate EU from CPT at all, and even those that are may have different beliefs about levels of risk aversion or probability distortion. We conduct two exercises which cast doubt on the possibility that these facts drive apparent sub-optimality. First, we inspect the design power distribution for the 7 alternative theory pairs from our experiment, comparing each to the design power of algorithmic designs optimized for that pair. For every single pair except for RCPT-REU, we observe a substantial gap between the best-powered literature design and the best-powered algorithmic design (see Figure 5a).

Second, we inspect the design power distribution compared to algorithmic design power under many possible parameter values for CPT. Recall that when we calculate $\bar{\rho}$, we are taking an average over a set of hypothesized parameter values determined by us. In principle, it could be the case that our algorithmic design does well on average; but worse for some particular parameter values. To assess this, we can separately calculate $\bar{\rho}$ for different hypothesized values of α and γ , both in the literature and on our algorithmic design. We report results in Table 5b. We see variation in the overall level of $\bar{\rho}$ depending on which parameter values we use – in particular, $\bar{\rho}$ is lower when γ is closer to 1, which reflects that less probability distortion make CPT more like EU. Despite these level differences, design power of the algorithmic designs is consistently higher than even the best existing design. We also find this reassuring because it indicates that even if researchers don’t select quite the right parameter values in the power objective, there are still benefits to optimizing.

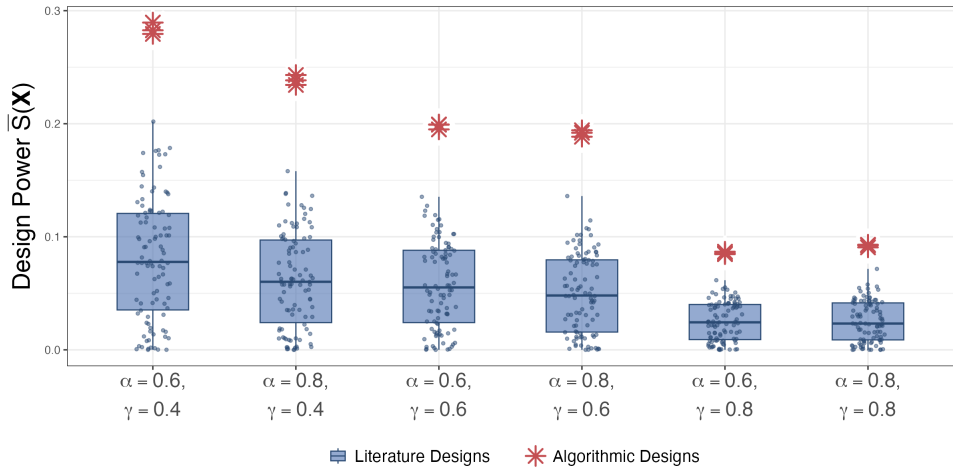
5.4 Connecting Power to Results

Given our empirical exercise in the previous section, a natural question is whether design power can help us understand any variation in results across different experimental work. Conducting the same exercise here is challenging because of the wide variation in model specifications and reported statistics across all of these papers. Some papers report log likelihoods, some report mean squared errors, and still others report rates of axiom violation.

Given these issues, we attempt a less formal exercise based on the text of the papers. First, we limit to the 65 papers which we classified as testing expected utility theory. Then,



(a) Average normalized separation for alternative theory pairs.



(b) CPT-EUT normalized separation for different CPT parameters.

Figure 5: Separation of literature designs and best algorithmic designs for different theory pairs (Panel a) and priors (Panel b)

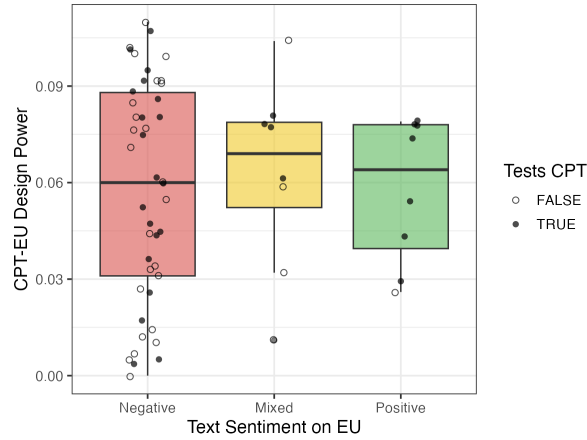


Figure 6: Design power for separating CPT from EUT, grouped by paper’s EUT sentiment.

we prompt an LLM to read the text of these papers and characterize whether the sentiment towards Expected Utility Theory is Positive, Negative, or Mixed. We exclude four papers with no clear sentiment. In Figure 6, we compare design power for separating CPT from EUT in each of the three groups.

If CPT were generally the reason EUT fails to explain data, we would expect to see a *negative* relationship between design power and EUT sentiment. Instead, we see no clear differences between groups. Still, considering these designs through the lens of power can help us interpret experimental results.

We make two observations. First, we notice that all of the papers which came to a positive conclusion about EUT are relatively low-powered (about 1/3 the value of our algorithmic design). This should make us take the success of EUT on these designs with a big grain of salt – even if CPT were right, we would expect EUT to fit the data well.

Second, we notice that many *very* low-powered designs deliver negative evidence for EUT. If our CPT-prior were the right alternative description of behavior, this wouldn’t happen – so what does this tell us? It tells us exactly that the violations of EUT observed on these designs are not what we would predict under CPT. Often this is apparent reading the papers: many don’t claim to test or study CPT at all, but instead focus on other non-EUT alternatives, or simply document a failure. Another way of putting this is that on many of these designs, the behavioral patterns that violate EUT would also violate CPT.

We can also learn something from the papers which explicitly study CPT, yet reject EUT with a low-powered design. To illustrate, we focus on one particular highly-studied lottery choice design: Allais’s common consequence problem (Allais, 1953). The design is composed of the two binary choices in Figure 7.

Choice 1

Option A: \$5 million with probability 10%, \$1 million with probability 89%, \$0 otherwise
Option B: \$1 million with probability 100%

Choice 2

Option C: \$5 million with probability 10%, \$0 otherwise
Option D: \$1 million with probability 11%, \$0 otherwise

Figure 7: Allais’s common consequence paradigm. The Allais paradox refers to the modal choice pattern of selecting Option B in Choice 1 and Option C in Choice 2, which violates expected utility.

The design is famous because it canonically illustrates a failure of EUT; under EUT preferences, a rational subject should either select *A* in Choice 1, *C* in Choice 2; or *B* in Choice 1, *D* in Choice 2. Instead, subjects often select *B* in Choice 1 and *D* in Choice 2. This paradox has been generative for theorists – many theories of non-EU risk preferences explicitly use it as a motivating example, including the seminal Tversky and Kahneman (1992) (KT) which first introduced Cumulative Prospect Theory.

If we return to Figure 4, we notice something puzzling: the classic Allais paradox, directly cited in Tversky and Kahneman (1992), is a very low-powered design for separating CPT from EU. How can this be true, when CPT was constructed precisely to explain a known breakage of EU on this pair of problems? It turns out that at all of the CPT parameter values we considered, guided by the later literature, CPT does *not* predict a breakage of EU. You might worry that this is hindsight bias – but we can also use the CPT parameters estimated by KT in this very paper. At these estimates ($\alpha = 0.88$, $\gamma = 0.61$), KT’s own model predicts a subject would prefer the risk-averse option in both choices.²⁰

Concretely, this means that CPT—as we know it—actually *does not* explain EU violations on the Allais Paradox. We are not the first to make this observation – this was also pointed out by Neilson and Stowe (2002). The disconnect comes from separately considering what the theory *can* accommodate, and what parameter values actually explain the data. Conceptually, CPT explains the failure of the independence axiom by positing that subjects overweight the 1% state in Choice 1. In order to actually get a preference flip, though, the degree of overweighting needs to be very large – much larger than the level we observe when we look at the collection of choices in KT’s experiment, or elsewhere in the risky choice literature. The incongruence here is obvious once we calculate power using the estimated parameters,

²⁰KT use power utility and a probability-weighting function of the form $w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}}$.

but goes unmentioned in the paper and likely unnoticed by the authors.

Zooming back out, these examples illustrate two broad types of inference errors that our framework for design power can help us avoid. First, we should be careful about interpreting “positive” evidence for a theory on a design which is low-powered to separate it from alternatives. And second, if we find negative evidence for a theory on a design we expected to be low-power relative to some alternative; this tells us we’re likely thinking of the wrong alternative.

6 Conclusion

In this paper, we have developed a framework for systematically approaching experimental design. We proposed a criterion for design based on statistical power for model selection and laid out an algorithmic procedure for generating highly-powered designs. We showed using choice under risk experiments that better designs according to the criterion tend to produce stronger statistical results. We also showed experimentally that asymmetries in design power can translate into predictable variation in statistical results, shifting them in favor of the asymmetrically-powered theory. Finally, we looked at the existing literature on choice under risk experiments. We observed that existing designs appear low-powered relative to designs discovered by our search algorithm. We looked specifically at designs which tested expected utility theory (EUT), and noted that the papers which came to favorable conclusions about the theory were generally low-powered to separate cumulative prospect theory (CPT) from EUT. We also noted that many *did* reject EUT despite being low-powered to separate CPT from EUT. Considering power on such designs is important when interpreting results, because it highlights that while EUT does fail, CPT is not a good descriptive theory of *why* EUT fails.

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A General Decision Theory

A.1 The General Decision Problem

The model-selection problem in the main text is one instance of a more general experimental design problem. Let $X \in \mathcal{X}_\ell^T$ be a feasible design and let $\mathbf{Y}$ denote the data it generates. The data-generating process is an unknown element p_0 of a set $\mathcal{P}$; a researcher may summarize prior knowledge about p_0 with a probability measure Π on $\mathcal{P}$. After observing the data, the researcher takes an action a from an action space $\mathcal{A}$ using a decision rule

$$\delta : \mathcal{Y}^{NT} \longrightarrow \mathcal{A}.$$

The loss from taking action a when the truth is p_0 is $\ell(a, p_0)$. Holding the decision rule fixed, the ex ante loss of design X is therefore

$$\mathcal{L}(X; \delta) = \int_{\mathcal{P}} \int \ell(\delta(\mathbf{Y}), p_0) p_0(d\mathbf{Y} | X) \Pi(dp_0). \quad (14)$$

The general design problem is to choose

$$X^* \in \arg \min_{X \in \mathcal{X}_\ell^T} \mathcal{L}(X; \delta).$$

The researcher must choose X before observing $\mathbf{Y}$, so this choice necessarily uses prior knowledge about which data-generating processes are reasonable candidates for p_0 . We let $\mathcal{P}$ denote the candidate models, and Π the weights the researcher assigns to each model in $\mathcal{P}$. We do not need to interpret Π as a Bayesian prior, or assume that the true model is literally drawn from $\mathcal{P}$ according to Π . Instead, we think of $\mathcal{P}$ and Π as a convenient way to translate approximate knowledge about the true data generating process into a concrete design criterion. Note that these objects are used to formulate our scientific goal (the loss function); but play no role in the statistical procedure used after data collection (the decision rule).

A.2 Model Selection

Suppose the researcher compares K model classes $\mathcal{M}_k = \{g_k(\cdot | \cdot, \theta_k) : \theta_k \in \Theta_k\}$. In this case, a natural candidate set is $\mathcal{P} = \bigcup_{k=1}^K \mathcal{M}_k$. The prior knowledge represented by Π can be read as a hierarchical Bayesian prior. First draw a model index k with probability P_k ,

then draw θ_k from a distribution π_k over Θ_k , and set $p_0(\cdot | X_t) = g_k(\cdot | X_t, \theta_k)$. The action is the selected model $\hat{k}$, and the loss $\ell(a, p_0) = -\mathbf{1}\{a = k\}$ rewards correct selection. In this special case, minimizing (14) is equivalent to maximizing the weighted-average probability of correct selection.

$$X^* \in \arg \max_{X \in \mathcal{X}_\ell^T} \sum_{k=1}^K P_k \int_{\Theta_k} \Pr(\hat{k} = k | X, (k, \theta_k)) \pi_k(d\theta_k).$$

The main text uses the same logic with a finite grid of plausible parameter values and researcher-specified weights.

A.3 GARP and Bronars' Power

Consider an experiment on revealed preference. A design $X = (X_1, \dots, X_T)$ specifies the sequence of budget lines. Subject i chooses a bundle on each line, producing the record $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{iT})$. Let $\mathcal{G}(X)$ be the set of records that satisfy the Generalized Axiom of Revealed Preference (GARP), so no revealed-preference cycle may contain a strict link. A test of GARP rejects when at least one observed record lies outside this set,

$$\delta(\mathbf{Y}_1, \dots, \mathbf{Y}_N) = \mathbf{1}\{\exists i \leq N : \mathbf{Y}_i \notin \mathcal{G}(X)\}.$$

To evaluate a design before collecting data, Bronars considers a simple benchmark data-generating process. For each candidate design X , let P_X^U denote the process under which every subject and every task draws a bundle uniformly along the corresponding budget line, independently across subjects and tasks. We can represent this benchmark in the general framework by taking $\mathcal{P}$ to contain these random-choice processes and by putting all weight on the uniform random-choice rule. This rule induces P_X^U for each candidate design X . Unlike the model-selection example, this is best understood as a convenient benchmark rather than a systematic Bayesian prior. Uniform random choice is a reference distribution used to compare designs, not a substantive claim about how preferences are generated. Bronars' power is the probability that the test rejects under this benchmark.

$$X^* \in \arg \max_{X \in \mathcal{X}_\ell^T} \Pr_{P_X^U}(\exists i \leq N : \mathbf{Y}_i \notin \mathcal{G}(X)). \quad (15)$$

Thus this example has the same structure as (14). The researcher specifies the action, the loss implicit in the testing goal, and a data-generating distribution that serves as the ex ante reference, then chooses the design with the best expected performance.

B Heterogeneity

Section 2 discusses the role of design power for model selection under the assumption that subjects are homogeneous. More broadly, researchers are often interested in heterogeneity across subjects and use experiments not only to select one model, but also to estimate the population share of people whose behavior is consistent with each model. In this section, we analyze common procedures used to estimate population shares, and show that the effect size $\bar{\rho}$ and the resulting power $\Phi(\sqrt{T} \bar{\rho})$ also govern the statistical properties of the estimators.

We first describe the data-generating process implicitly assumed in many papers that analyze heterogeneity. We then study two procedures used in practice. Our goal here is not to match every empirical implementation exactly, but to capture the role of design power in a broad class of applications.

We assume the data generating process is as follows. There are N subjects and T tasks. Subject i has an unobserved type $Z_i \in \mathcal{K} = \{1, \dots, K\}$, with population shares w^0 , and a type-specific parameter $\theta_i \mid Z_i = k \sim F_k(\cdot \mid \eta_k^0)$. Conditional on $(Z_i = k, \theta_i)$, outcomes are independent across tasks with $Y_{it} \sim g_k(\cdot \mid X_t, \theta_i)$. The point-mass case $F_k = \delta_{\theta_k}$ is the finite mixture where subjects within a type share a common set of parameters, whereas a nondegenerate F_k allows parameters to vary within type.

The first procedure is *individual classification*. The researcher assigns each subject to a type, either by estimating the subject’s best-fitting model or by applying a prespecified rule such as “the subject violates the model’s prediction.” She then counts the assignments to estimate type shares (Choi et al., 2007; Cavagnaro et al., 2013). The second procedure is *mixture estimation*, in which the researcher treats types as latent and estimates population shares jointly with the model parameters, typically by maximum likelihood (e.g., Barseghyan et al., 2021). For both procedures, the estimand is the population share w_k^0 of each type. We next characterize how design power affects these estimators.

B.1 Individual classification

To illustrate the first procedure formally, consider the case where the individual-level classification is based on likelihood comparisons.

$$\begin{aligned}\hat{\theta}_{k,i} &\in \arg \max_{\theta \in \Theta_k} \sum_{t=1}^T \log g_k(Y_{it} | X_t, \theta), \\ \hat{Z}_i &\in \arg \max_{k \in \mathcal{K}} \sum_{t=1}^T \log g_k(Y_{it} | X_t, \hat{\theta}_{k,i}),\end{aligned}\quad \hat{w}_k^{\text{cls}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{Z}_i = k\}.$$

For $K = 2$, define subject i 's fitted average likelihood advantage of class k over the other class as

$$\hat{D}_{ik}^{\text{cls}} = \frac{1}{T} \sum_{t=1}^T \log \frac{g_k(Y_{it} | X_t, \hat{\theta}_{k,i})}{g_{3-k}(Y_{it} | X_t, \hat{\theta}_{3-k,i})}.$$

Define the confusion matrix by

$$Q_{kk'} = \Pr(\hat{Z}_i = k' | Z_i = k),$$

where the probability also averages over $\theta_i \sim F_k(\cdot | \eta_k^0)$. The law of large numbers gives the following basic result.

Proposition 2 (Individual classification).

For any number of classes K and fixed T , as $N \rightarrow \infty$,

$$\hat{w}^{\text{cls}} \xrightarrow{p} Q^\top w^0.$$

For each class, writing $r = Q^\top w^0$, the share estimator has

$$\text{MSE}(\hat{w}_k^{\text{cls}}) = \frac{r_k(1-r_k)}{N} + (r_k - w_k^0)^2. \quad (16)$$

For $K = 2$, under analogous likelihood regularity and task-level Lindeberg conditions, as $T \rightarrow \infty$,

$$\frac{\sqrt{T}(\hat{D}_{ik}^{\text{cls}} - \bar{D}_{k,3-k}(\theta; X))}{\sqrt{\bar{V}_{k,3-k}(\theta; X)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

conditional on $Z_i = k$ and $\theta_i = \theta$. Consequently, the Gaussian approximation to the diagonal

classification probabilities is

$$Q_{kk} \approx \mathbb{E}_{\theta \sim F_k} [\Phi(\sqrt{T} \bar{\rho}_{k,3-k}(\theta; X))], \quad Q_{k,3-k} = 1 - Q_{kk}. \quad (17)$$

Proposition 5 indicates that the population share estimator $\hat{w}^{\text{cls}}$ converges not to the true shares of types, but instead to the shares of assigned labels. If the individual-level classification is imperfect, then these two shares do not coincide. The confusion matrix characterizes the quality of individual-level classification, and its diagonal entries denote the probability of correct classification for each type. The two shares are close only when the classification matrix is close to the identity. In the two-class case, the Gaussian approximation in (28) shows how this matrix is governed by the per-subject effect size and by T , rather than by NT . More subjects reduce the sampling variance in (29), but cannot reduce the asymptotic bias originating from uncertainty in the individual classification step. Low-power designs can therefore distort estimated shares regardless of how many subjects the experiment includes.

The proof is in Appendix D.4.

B.2 Mixture estimation

The second procedure estimates shares without assigning a single type to each subject. Instead, each subject receives a probability of belonging to each type, and these latent assignments contribute to the likelihood for the population shares. A common implementation estimates the weights and one within-type parameter for each class jointly:

$$(\hat{w}^{\text{mix}}, \hat{\theta}) \in \arg \max_{w \in \Delta^{K-1}, \tilde{\theta}_1, \dots, \tilde{\theta}_K} \sum_{i=1}^N \log \left[\sum_{k=1}^K w_k \prod_{t=1}^T g_k(Y_{it} | X_t, \tilde{\theta}_k) \right]. \quad (18)$$

This procedure uses each subject's full record but imposes a common parameter within each type. When the true distributions F_k are point masses, this is the correctly specified finite mixture. However, with parameter heterogeneity, it is generally misspecified. Let $(w^*, \tilde{\theta}^*)$ denote the pseudo-true joint maximizer of the population version of (18). We suppress its dependence on X to simplify notation. For subject i , define the posterior probability of type k' as

$$\tau_{ik'}^* = \frac{w_{k'}^* \prod_t g_{k'}(Y_{it} | X_t, \tilde{\theta}_{k'}^*)}{\sum_j w_j^* \prod_t g_j(Y_{it} | X_t, \tilde{\theta}_j^*)},$$

and the soft confusion matrix

$$Q_{kk'}^* = \mathbb{E}[\tau_{ik'}^* | Z_i = k],$$

where the expectation also averages over $\theta_i \sim F_k$.

For two classes, the connection to design power follows from the posterior log odds. To see this, we define

$$A_{ik}^* \equiv \log \frac{\tau_{ik}^*}{\tau_{i,3-k}^*} = \log \frac{w_k^*}{w_{3-k}^*} + \sum_{t=1}^T \log \frac{g_k(Y_{it} | X_t, \tilde{\theta}_k^*)}{g_{3-k}(Y_{it} | X_t, \tilde{\theta}_{3-k}^*)}, \quad (19)$$

then conditional on a type- k subject with parameter θ , let $h(a) = (1 + e^{-a})^{-1}$ and denote the scale and effect size of the posterior log odds by

$$\begin{aligned} \bar{\sigma}_k^{\text{mix}}(\theta; X) &= \sqrt{T^{-1} \text{Var}(A_{ik}^* | Z_i = k, \theta_i = \theta)}, \\ \bar{\rho}_k^{\text{mix}}(\theta; X) &= \frac{T^{-1} \mathbb{E}[A_{ik}^* | Z_i = k, \theta_i = \theta]}{\bar{\sigma}_k^{\text{mix}}(\theta; X)}. \end{aligned} \quad (20)$$

The numerator has a direct KL interpretation. Under a type- k subject with parameter θ ,

$$\begin{aligned} \frac{1}{T} \mathbb{E}[A_{ik}^* | Z_i = k, \theta_i = \theta] &= \frac{1}{T} \log \frac{w_k^*}{w_{3-k}^*} \\ &\quad + \frac{1}{T} \sum_{t=1}^T \text{KL}(g_k(\cdot | X_t, \theta) \parallel g_{3-k}(\cdot | X_t, \tilde{\theta}_{3-k}^*)) \\ &\quad - \frac{1}{T} \sum_{t=1}^T \text{KL}(g_k(\cdot | X_t, \theta) \parallel g_k(\cdot | X_t, \tilde{\theta}_k^*)). \end{aligned} \quad (21)$$

The first KL term measures separation from the fitted alternative while the second term subtracts the error from representing a heterogeneous type- k subject with the common parameter $\tilde{\theta}_k^*$. With correctly specified point-mass types, the second term is zero, so this is the original KL separation $\bar{D}_k$ plus the weight-odds term divided by T .

The Gaussian approximation can then be written directly in terms of the effect size,

$$\tau_{ik}^* = h(A_{ik}^*) \approx h(\sqrt{T} \bar{\sigma}_k^{\text{mix}}(\theta; X) [\sqrt{T} \bar{\rho}_k^{\text{mix}}(\theta; X) + Z]), \quad Z \sim \mathcal{N}(0, 1). \quad (22)$$

Equation (22) makes the role of design power explicit. It uses the same signal-to-noise normalization as $\bar{\rho}_k$ in the main text. With correctly specified point-mass types, it reduces to the original effect size, apart from the mixture-weight log odds divided by T . With parame-

ter heterogeneity, its numerator is the KL distance to the fitted rival minus the KL distance to the fitted version of the subject's own class. Mixture estimation makes soft rather than hard assignments, so the effect size enters through the logistic function. Holding $\bar{\sigma}_k^{\text{mix}}$ fixed, higher $\bar{\rho}_k^{\text{mix}}$ moves posterior weight toward the true type and reduces cross-type confusion. Unlike the hard assignment probability $\Phi(\sqrt{T} \bar{\rho}_k^{\text{mix}})$, however, the soft assignment depends on both the effect size and the scale of the log odds.

Proposition 3 (Mixture estimation with common within-type parameters).

Under the usual regularity conditions for maximum likelihood,

$$\hat{w}^{\text{mix}} \xrightarrow{P} w^* = Q^{*\top} w^0, \quad \sqrt{N}(\hat{w}^{\text{mix}} - w^*) \xrightarrow{d} \mathcal{N}(0, V_X).$$

Consequently, for each class k , the large- N mean squared error is

$$\text{MSE}(\hat{w}_k^{\text{mix}}) \approx \frac{(V_X)_{kk}}{N} + (w_k^* - w_k^0)^2.$$

The proof is in Appendix D.5.

If F_k is a point mass at the true parameter for every k , the mixture is correctly specified. The first-order condition for the weights then gives $Q^{*\top} w^0 = w^0$, even though individual posterior probabilities need not be degenerate. Thus, design power affects sampling variance but not the probability limit. If the true F_k is nondegenerate while the estimator imposes a common parameter within each type, the mixture is misspecified and we generally have $w^* \neq w^0$. Design separation can then affect both asymptotic bias and sampling variance. The intuition for the bias is similar to the previous subsection, which is that under a misspecified model, individual classification is poor when design power is low, and thus bias is introduced to the estimation of aggregate shares. Similarly, this bias cannot be eliminated by increasing the number of subjects.

C Choosing a Design with More Than Two Model Classes

Section 2.2 presents our design criterion for two model classes. When there are $K > 2$ classes, a fixed generating model must be distinguished from $K - 1$ rivals, hence its normalized separation $\bar{\rho}_k(\theta_k; \mathbf{X}) = (\bar{\rho}_{kj}(\theta_k; \mathbf{X}))_{j \neq k}$ is a vector with one entry for each rival model class. Since we prefer a scalar design criterion, we first need to turn each $\bar{\rho}_k(\theta_k; \mathbf{X})$ into a scalar by summarizing the pairwise separations.

Our preferred summary is the minimum:

$$\bar{\rho}_k^{\min}(\theta_k; \mathbf{X}) = \min_{j \neq k} \bar{\rho}_{kj}(\theta_k; \mathbf{X}). \quad (23)$$

This is the normalized separation from the rival that is hardest to distinguish from (M_k, θ_k) on design $\mathbf{X}$. The minimum is a natural criterion when the goal is to select M_k over all of its rivals: if one rival remains difficult to distinguish, large separations from the other classes do not resolve that comparison.²¹

The rest of the design recipe follows Section 2.2. For each class k , the researcher specifies a parameter grid $(\theta_{ks})_{s=1}^{S_k}$ with nonnegative weights (w_{ks}) that sum to one, as well as nonnegative model weights $(P_k)_{k=1}^K$ that sum to one. The resulting criterion is

$$\mathbf{X}^* \in \arg \max_{\mathbf{X} \in \mathcal{X}_\ell^T} \sum_{k=1}^K P_k \sum_{s=1}^{S_k} w_{ks} \bar{\rho}_k^{\min}(\theta_{ks}; \mathbf{X}). \quad (24)$$

When $K = 2$, the expression in (24) reduces to (7).

We note that taking the minimum in (23) is a simplification of the vector separation. The minimum does not recover the full power expression in Proposition 1, since it also depends on how the likelihood comparisons against different rivals co-move. The following proposition shows, however, that the minimum bounds approximate power without requiring these correlations.

Proposition 4 (Power bounds with K classes).

Under the conditions of Proposition 1, fix a model (M_k, θ_k) and a design $\mathbf{X}$, and let $n = NT$ and $m = \bar{\rho}_k^{\min}(\theta_k; \mathbf{X})$. For any correlation matrix $\bar{R}_k$,

$$1 - (K - 1) \Phi(-\sqrt{n}m) \leq \Phi_{\bar{R}_k}(\sqrt{n} \bar{\rho}_k(\theta_k; \mathbf{X})) \leq \Phi(\sqrt{n}m), \quad (25)$$

and the upper bound is attained when the standardized evidence against all rivals is perfectly positively correlated. Consequently, if $\bar{\rho}_k^{\min}(\theta_k; \mathbf{X}_A) > \bar{\rho}_k^{\min}(\theta_k; \mathbf{X}_B)$, then $\mathbf{X}_A$ has higher approximate power than $\mathbf{X}_B$ for all sufficiently large n , regardless of the correlation structures under the two designs.

The proof is in Appendix D.6. To interpret the result, let $q = \Phi(-\sqrt{n}m)$ denote the Gaussian approximation to the probability that the hardest-to-distinguish rival beats M_k .

²¹Here the minimum is taken across rival classes for a fixed generating model. This differs from the symmetric objective $\min\{\bar{\rho}_1, \bar{\rho}_2\}$ in Section 4, which takes the minimum across two possible generating models.

Within this approximation, the probability of selecting the wrong class lies between q and $(K - 1)q$. Thus $\bar{\rho}_k^{\min}$ determines the rate at which the approximate error probability falls with sample size, up to a factor of at most $K - 1$. Correlation across comparisons determines where power lies within these bounds, but it cannot overturn the eventual ranking of two designs whose minimum separations are strictly ordered.

We have used the minimum separation $\bar{\rho}_k^{\min}$, but one could instead work with average separation $\bar{\rho}_k^{\text{avg}} = (K - 1)^{-1} \sum_{j \neq k} \bar{\rho}_{kj}$. Because $\bar{\rho}_k^{\text{avg}} \geq \bar{\rho}_k^{\min}$, $\Phi(\sqrt{n} \bar{\rho}_k^{\text{avg}})$ is also an upper bound on approximate power, but it is not in general a lower bound. Intuitively, we can have a design where one rival is very challenging to separate, but others are very easy. Modifying the design to increase separation from the “easy-to-distinguish” rivals increases average power; but does not necessarily improve the comparison limiting our probability of correct selection. Since we have formulated our design goal as maximizing the probability of correct selection, we prefer the minimum; a researcher focused on improving a typical pairwise comparison may instead prefer the average. Under the decision-theoretic formulation in Appendix A.1, neither object exactly matches the expected-loss criterion, which uses the multivariate probability of correct selection. Nonetheless, the minimum is a natural scalar proxy which (1) focuses on the hardest comparison and (2) does not require the researcher to specify the correlation structure across all pairwise comparisons.

D Proofs

D.1 Proof of Proposition 1

We give the argument for a fixed design and a fixed scenario (k, θ_k) ; write $n = NT$. Let q_j^* denote the pseudo-true distribution in class j on the chosen tasks, let θ_j^* be any parameter inducing q_j^* , and let $\hat{\ell}_j$ be the maximized log likelihood per observation. The likelihood regularity condition in Proposition 1(c) is that, jointly over the finitely many model classes,

$$\hat{\ell}_j = \frac{1}{n} \sum_{i=1}^N \sum_{t=1}^T \log g_j(Y_{it} | X_t, \theta_j^*) + o_p(n^{-1/2}), \quad (26)$$

where $\theta_k^* = \theta_k$ for the true class. In words, maximizing over a model’s parameters has no first-order effect relative to evaluating its likelihood at a pseudo-true distribution.

For an ordinary finite-dimensional smooth likelihood model, Equation (26) follows from the usual consistency and local quadratic expansion of the likelihood; under standard interiority and nonsingularity conditions, the remainder is in fact $O_p(n^{-1})$; see ?, Chapter 5,

especially Sections 5.2–5.6. The condition also accommodates a model class that is a finite union of smooth components. A sufficient set of conditions is that all KL-minimizing components induce the same distribution on the chosen tasks, every other component has a strictly positive population-likelihood gap, and the usual likelihood expansion holds within each minimizing component. Uniform convergence then excludes the non-minimizing components with probability approaching one, while Proposition 1(a) ensures that whichever minimizing component is selected has the same first-order likelihood target. In the pure-tremble model in the main text, the preference parameter partitions a fixed design into finitely many choice patterns, so this finite-union argument applies (with the usual smooth likelihood conditions within a pattern if the tremble probability is estimated).

For each subject define the vector of log-likelihood ratios against the projected rivals,

$$\Lambda_i = \left(\frac{1}{T} \sum_{t=1}^T \log \frac{g_k(Y_{it} | X_t, \theta_k)}{g_j(Y_{it} | X_t, \theta_j^*(\theta_k, X))} \right)_{j \neq k}.$$

The vectors Λ_i are i.i.d. across subjects. By Definition 2,

$$\mathbb{E}[\Lambda_i] = \bar{\mathbf{D}}_k, \quad \text{Var}(\Lambda_i) = \frac{1}{T} \bar{\Sigma}_k, \quad (\bar{\Sigma}_k)_{jj'} = \frac{1}{T} \sum_{t=1}^T \text{Cov}(L_t^{kj}, L_t^{kj'}).$$

The multivariate central limit theorem therefore gives

$$\sqrt{NT} (\bar{\Lambda} - \bar{\mathbf{D}}_k) \xrightarrow{d} \mathcal{N}(0, \bar{\Sigma}_k), \quad \bar{\Lambda} = \frac{1}{N} \sum_{i=1}^N \Lambda_i. \quad (27)$$

By (26), the vector of fitted likelihood differences satisfies

$$\hat{\mathbf{D}}_k = \bar{\Lambda} + o_p(n^{-1/2}).$$

Combining this expansion with (27) gives

$$\sqrt{NT} (\hat{\mathbf{D}}_k - \bar{\mathbf{D}}_k) \xrightarrow{d} \mathcal{N}(0, \bar{\Sigma}_k),$$

which proves Proposition 1.

Let $D_k = \text{diag}(\sqrt{\bar{V}_{kj}})_{j \neq k}$. Then $D_k^{-1} \bar{\mathbf{D}}_k = \bar{\rho}_k$ and $D_k^{-1} \bar{\Sigma}_k D_k^{-1} = \bar{R}_k$. Standardizing (27) and using symmetry of a centered Gaussian vector gives the Gaussian power approximation

reported in the main text:

$$\Pr(\hat{k} = k \mid \mathbf{X}, (k, \theta_k)) \approx \Pr(W \leq \sqrt{NT} \bar{\rho}_k(\theta_k; \mathbf{X})) = \Phi_{\bar{R}_k}(\sqrt{NT} \bar{\rho}_k(\theta_k; \mathbf{X})),$$

where $W \sim \mathcal{N}(0, \bar{R}_k)$. For $K = 2$, this is the scalar probability $\Phi(\sqrt{NT} \bar{\rho}_k)$. If a total-log-likelihood penalty $p_{j,n}$ is subtracted from class j 's fitted likelihood, the penalized average gap differs from $\hat{D}_{kj}$ by $(p_{k,n} - p_{j,n})/n$. Hence differences in penalties that are $o(\sqrt{n})$ are $o(n^{-1/2})$ on the average-likelihood scale and do not change the first-order distribution or the resulting power approximation.

D.2 Proof of Lemma 1

Write $r_A = \bar{\rho}(X_A)$, $r_B = \bar{\rho}(X_B)$, and $g = 1 - r_B/r_A$. For $x \geq 0$, the standard normal CDF satisfies

$$\Phi(x) - \frac{1}{2} \leq \phi(0)x, \quad \Phi(x) - \frac{1}{2} \geq \phi(0)x \left(1 - \frac{x^2}{6}\right).$$

The second inequality follows by integrating the bound $e^{-u^2/2} \geq 1 - u^2/2$; the first follows from concavity of Φ on $[0, \infty)$. By assumption, every scenario-specific effect size under X_A satisfies $\sqrt{n} \bar{\rho}_s(X_A) < z_\beta$. The gap condition implies $z_\beta^2 \leq 6g$, so the lower bound applied scenario by scenario to X_A and the upper bound applied to X_B yield

$$\text{WAP}(X_A; n) - \frac{1}{2} \geq \phi(0)\sqrt{n} r_A(1 - g) = \phi(0)\sqrt{n} r_B \geq \text{WAP}(X_B; n) - \frac{1}{2}.$$

The first inequality is strict because $r_A > 0$ and the separation condition is strict. Hence $\text{WAP}(X_A; n) > \text{WAP}(X_B; n)$.

D.3 Proof of Corollary 1

Consider first truth $(1, \theta_1)$. The proof of Proposition 1 shows, for two models, that

$$\hat{D} = \bar{D}_{12}(\theta_1; X) + O_p(n^{-1/2}).$$

Finite second moments and a positive separated variance imply that the sample variance used in $\hat{\sigma}^2$ converges to $\bar{V}_{12}(\theta_1; X)$. Therefore

$$\hat{z} = \sqrt{n} \bar{\rho}_{12}(\theta_1; X) + O_p(1).$$

Under truth $(2, \theta_2)$, reversing the likelihood ratio changes both centers' signs, giving the first part of the corollary.

For the second part, integrate these conditional limits over the model and scenario weights. The assumed moment bounds give uniform integrability, so the limits can be taken through expectations. Under truth 1, the signed mean is $\mathbb{D}_1$ and the absolute mean is also $\mathbb{D}_1$; under truth 2, the signed mean is $-\mathbb{D}_2$ and the absolute mean is $\mathbb{D}_2$. The same argument applied to $\hat{z}/\sqrt{n}$ gives $\bar{\rho}_1$ and $-\bar{\rho}_2$. Weighting the two cases by P_1 and P_2 gives the four limits in Corollary 1.

D.4 Proof of Proposition 5

Holding T fixed, the assignment indicators are i.i.d. across subjects, with $\Pr(\hat{Z}_i = k') = (Q^\top w^0)_{k'}$. The law of large numbers therefore gives $\hat{w}^{\text{cls}} \rightarrow_p Q^\top w^0$. For each class k , the indicators $\mathbf{1}\{\hat{Z}_i = k\}$ are Bernoulli with success probability $r_k = (Q^\top w^0)_k$, so

$$\text{MSE}(\hat{w}_k^{\text{cls}}) = \frac{r_k(1-r_k)}{N} + (r_k - w_k^0)^2.$$

For the second result, fix a type k and parameter θ . Under the stated likelihood regularity conditions, maximizing each within-subject likelihood has no first-order effect relative to evaluating it at its pseudo-true distribution, so

$$\hat{D}_{ik}^{\text{cls}} = \frac{1}{T} \sum_{t=1}^T L_t^{k,3-k} + o_p(T^{-1/2}).$$

The task-level log-likelihood ratios have total mean and variance, respectively,

$$T\bar{D}_{k,3-k}(\theta; X) \quad \text{and} \quad T\bar{V}_{k,3-k}(\theta; X).$$

The Lindeberg central limit theorem and the preceding likelihood expansion therefore give

$$\frac{\sqrt{T}(\hat{D}_{ik}^{\text{cls}} - \bar{D}_{k,3-k}(\theta; X))}{\sqrt{\bar{V}_{k,3-k}(\theta; X)}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Because the classifier assigns the subject to class k when $\hat{D}_{ik}^{\text{cls}} \geq 0$, the corresponding Gaussian approximation is

$$\Pr(\hat{Z}_i = k \mid Z_i = k, \theta_i = \theta) \approx \Phi(\sqrt{T} \bar{\rho}_{k,3-k}(\theta; X)).$$

Averaging over $\theta \sim F_k$ gives (28); for two classes, the off-diagonal probability is its complement.

D.5 Proof of Proposition 3

Let $\psi = (w, \tilde{\theta})$ collect the weights and common within-type parameters. The joint mixture estimator is an ordinary, possibly misspecified, M-estimator based on independent subject records. The law of large numbers gives $\hat{\psi} \rightarrow_p \psi^* = (w^*, \tilde{\theta}^*)$. The usual score expansion and subject-level central limit theorem give asymptotic normality; V_X is the weight block of the resulting sandwich covariance matrix. For the weights, the population score (using a Lagrange multiplier for the simplex constraint) is equivalent to

$$w_{k'}^* = \mathbb{E}[\tau_{ik'}^*] = \sum_{k=1}^K w_k^0 Q_{kk'}^*,$$

which gives $w^* = Q^{*\top} w^0$. If the mixture is correctly specified, the posterior probabilities satisfy the usual iterated-expectations identity, so the right-hand side equals w^0 . Finally,

$$\mathbb{E}[(\hat{w}_k^{\text{mix}} - w_k^0)^2] = \text{Var}(\hat{w}_k^{\text{mix}}) + (\mathbb{E}[\hat{w}_k^{\text{mix}}] - w_k^0)^2 \approx \frac{(V_X)_{kk}}{N} + (w_k^* - w_k^0)^2,$$

which is the MSE expression in Proposition 3.

D.6 Proof of Proposition 4

Let $W \sim \mathcal{N}(0, \bar{R}_k)$ and $v = \sqrt{n} \bar{\rho}_k(\theta_k; \mathbf{X})$, so that $\Phi_{\bar{R}_k}(v) = \Pr(W_j \leq v_j \text{ for all } j \neq k)$ and each W_j is standard normal. The event $\{W_j \leq v_j \text{ for all } j \neq k\}$ is contained in $\{W_{j'} \leq v_{j'}\}$ for the rival j' attaining $\min_j v_j = \sqrt{n} m$, which gives the upper bound. The union bound gives

$$\Pr(W_j \leq v_j \text{ for all } j \neq k) \geq 1 - \sum_{j \neq k} \Phi(-v_j) \geq 1 - (K-1) \Phi(-\sqrt{n} m),$$

which is the lower bound. If every entry of $\bar{R}_k$ equals one, all W_j coincide almost surely, so $\Pr(W_j \leq v_j \text{ for all } j \neq k) = \Phi(\min_j v_j) = \Phi(\sqrt{n} m)$ and the upper bound is attained. This is the case when the log-likelihood ratios L_t^{kj} coincide across j , for instance when every rival's pseudo-true model induces the same distribution on the tasks in $\mathbf{X}$.

For the last claim, write m_A and m_B for the two minima, with $m_A > m_B \geq 0$. Applying the lower bound to $\mathbf{X}_A$ and the upper bound to $\mathbf{X}_B$, the difference in approximate

power is at least $\Phi(-\sqrt{n} m_B) - (K - 1) \Phi(-\sqrt{n} m_A)$, which is positive for all large n because $\Phi(-\sqrt{n} m_A)/\Phi(-\sqrt{n} m_B) \rightarrow 0$.

E Algorithmic Design

Let $f : \mathcal{X}_t^T \rightarrow \mathbb{R}$ denote the objective in Equation (7), taken as a deterministic function of the design space $\mathbf{X}$. Our implementation follows the TuRBO algorithm of Eriksson et al. (2019), up to the minor implementation changes. To begin, we rescale the design space coordinates to the unit hypercube $[0, 1]^d$ and maintain a single adaptive trust region.

The main distinction between TuRBO and conventional Bayesian Optimization is that local trust regions are maintained in the optimization process to avoid difficulties caused by global idiosyncrasies of the objective function. Within the trust region, we fit a Gaussian-process (GP) model using only evaluations made since the region was initialized. In a GP, f is treated as an unknown function which is modeled probabilistically. When the GP learns the value of f at specific points (designs), it uses this information to form a predictive distribution of f at nearby points which have not been evaluated. We use a Matérn-5/2 covariance function and separate length scales for each design dimension. We also standardize f prior to fitting the GP. Given observations $\mathbf{y} = (y_1, \dots, y_t)^\top$, let K be their covariance matrix and let $\mathbf{k}(\mathbf{X})$ contain their covariances with a new design $\mathbf{X}$. The standard GP posterior has mean and covariance

$$\begin{aligned}\mu_t(\mathbf{X}) &= m + \mathbf{k}(\mathbf{X})^\top (K + \lambda I)^{-1} (\mathbf{y} - m \mathbf{1}), \\ k_t^{\text{post}}(\mathbf{X}, \mathbf{X}') &= k(\mathbf{X}, \mathbf{X}') - \mathbf{k}(\mathbf{X})^\top (K + \lambda I)^{-1} \mathbf{k}(\mathbf{X}'),\end{aligned}$$

where m is a fitted constant and λ is a small noise term used to stabilize the fit. The posterior therefore provides both a predicted objective value and uncertainty about that prediction.

The trust region is a hyperrectangle centered at the best design within it. If L is its base side length and ℓ_j is the GP length scale for coordinate j , its side length in that coordinate, before intersecting the region with $[0, 1]^d$, is $L_j = L \ell_j / (\prod_{h=1}^d \ell_h)^{1/d}$. Within this region, we generate candidate designs by changing each coordinate of the center with probability $\min\{1, 20/d\}$, while requiring every candidate to change at least one coordinate. To select a batch, we draw candidate values jointly from the GP posterior and select the candidate with the largest value in each draw, without selecting a candidate twice. Again, this form of Thompson sampling favors designs that either have high predicted values or remain sufficiently uncertain, striking a balance between exploration and exploitation.

The base side length begins at $L = 0.8$. It is doubled after consecutive successful batches and halved after consecutive unsuccessful batches, where success means improving the region’s best value by more than 10^{-3} times its absolute value. If the region persistently see no success and is shrunk to $L < 2^{-7}$, then it is terminated, and we draw a new Latin-hypercube initialization over the full design space. The GP then uses only data from the new region, while the best design found across all regions is retained.²²

To assess the performance of our preferred algorithm, we conduct a computational benchmark on two objectives: one-sided CPT truth against a pseudo-true EU alternative, and random-preference CPT (RP-CPT) truth against a pseudo-true random-preference EU (RP-EU) alternative. We consider designs with $T \in \{2, 4, 8, 12\}$ two-state lottery-choice tasks. Each task contributes five continuous design coordinates, so the search dimension ranges from 10 to 60. The objective averages the one-sided normalized separation $\bar{\rho}$ over six parameter values from the maintained prior for the true model.

We compare the performance of four methods after the same initializing grid evaluation, including (i) our reference-style TuRBO-1 implementation; (ii) a single BFGS search in logit-transformed coordinates²³, initialized at the best design from the common initialization; (iii) Nelder–Mead in logit-transformed coordinates, using ten starts selected from the best designs in the common initialization; and (iv) randomized maximin Latin-hypercube (LHS) search.²⁴

All benchmark methods search over continuous, common-state-space lotteries. Probabilities lie in $[0.05, 0.95]$ and payoffs lie in $[0, 100]$. Within each task, we parameterize payoffs to make lottery A weakly better in one state and lottery B weakly better in the other, thereby excluding strict dominance. Within each model-pair, task-count, and seed cell, all methods receive the same deterministic 100-point LHS initialization. This initialization is included in a maximum total budget of 1,600 objective evaluations. TuRBO uses batches of $q = 8$ and $C = 500$ candidates per batch. The inner pseudo-true-parameter calculation uses five baseline starting values together with deterministic boundary starts, for a total of seven starts for EU and nine for RP-EU.²⁵ We use the finalized parameter bounds except for CPT versus EU at $T = 2$, where the EU search is widened to $\alpha \geq 0.05$ and $\eta \leq 50$. When a

²²Our implementation differs from the original Eriksson et al. (2019) in that we use a randomized Latin hypercube rather than a scrambled Sobol sequence to generate candidates, and we use a smaller number of $C = 500$ candidates. We use GauPro to fit the GP.

²³We use logit-transformation so that BFGS can search over an unconstrained space instead of bounded within $[0, 1]$.

²⁴For our design space, a literal grid search is practically infeasible since the dimensionality is high, hence we use maximin LHS as the grid-like benchmark.

²⁵This is to avoid the inner pseudo-true-parameter optimizer getting stuck in local solutions.

local optimizer terminates before exhausting its allocation, we start a new round and count toward the 1,600-evaluation horizon.

Figure 8 summarizes the results. TuRBO-1 has the highest mean final objective among the displayed methods in every $T \geq 4$ comparison, with its advantage becoming especially pronounced as the parameter dimension increases. For smaller designs at $T = 2$, all four methods essentially approach the same ceiling. Therefore our computation suggests that TuRBO-1 performs well for the higher-dimensional problems that motivate our algorithm.

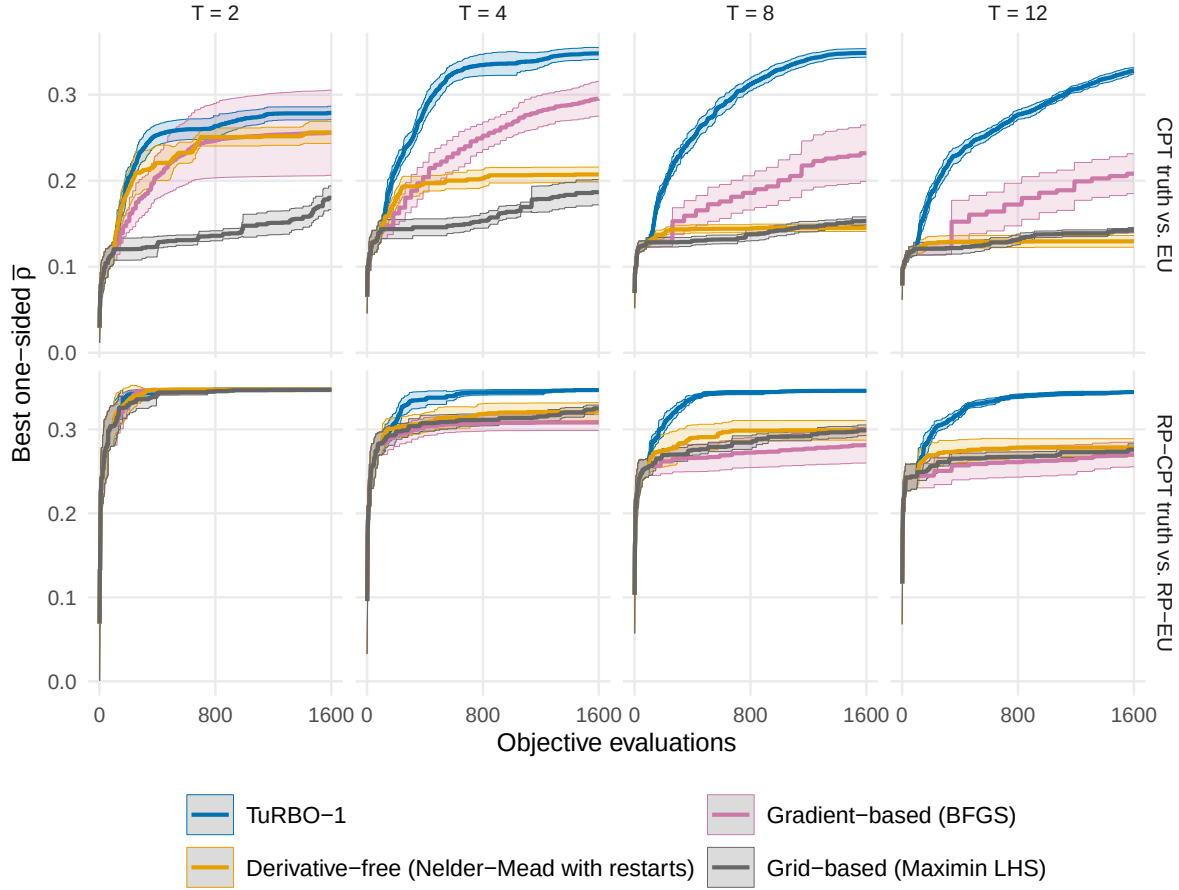


Figure 8: Optimizer convergence on the paper’s one-sided normalized-separation objective. Curves show the mean incumbent and shading gives pointwise 95% t confidence intervals for the mean across twelve seeds. The four displayed methods represent Bayesian optimization, gradient-based, derivative-free, and space-filling search.

F Robustness to Vuong-Statistic Degeneracy

The classical Vuong statistic used in Table 3 can sometimes be degenerate when fitted model classes overlap and are equally close to the realized empirical distribution. This changes the relevant distribution of the statistic and hence affects interpretation of magnitudes. In particular, comparing magnitudes of the statistics between degenerate and non-degenerate typically does not give rise to any clean interpretation. To protect against such issues we repeat the two specifications in Table 3 that use the Vuong statistic as the empirical outcome and replace it with the one-step non-degenerate statistic of Shi (2015). This statistic is modified based on the Vuong statistic and designed to control size uniformly in both overlapping and non-overlapping cases. For each model pair, we regress $|\hat{t}_{\text{ndV}}|$ on $\sqrt{NT} \bar{\rho}$; for the six non-nested pairs, we also regress $\hat{t}_{\text{ndV}}$ on $\sqrt{NT} \bar{\rho}_{\Delta}$. All 480 design-model-pair cells yield a finite statistic. The design sample, regressors, and transformations otherwise match Table 3.

Theory pair	$ \hat{t}_{\text{ndV}} \sim \sqrt{NT} \bar{\rho}$	$\hat{t}_{\text{ndV}} \sim \sqrt{NT} \bar{\rho}_{\Delta}$
CPT vs. EU	0.69*** (0.20)	
RCPT vs. REU	0.37*** (0.07)	
CC vs. CPT	0.48*** (0.09)	1.07*** (0.16)
CC vs. RCPT	0.12 (0.15)	0.62*** (0.18)
CC vs. SAL	0.63*** (0.11)	1.14*** (0.12)
CPT vs. RCPT	0.10 (0.15)	0.67* (0.36)
CPT vs. SAL	0.55** (0.24)	0.51 (0.33)
RCPT vs. SAL	0.13 (0.20)	0.83** (0.40)

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$

Table 4: Regressions corresponding to columns (2) and (4) of Table 3, using the non-degenerate Vuong statistic of Shi (2015) as the empirical outcome. Each regression contains all 60 designs for the indicated model pair. Parentheses report OLS standard errors; significance stars refer to tests that the regression slope equals zero. Directional regressions are not defined for the two nested pairs under our one-sided design objective.

All estimated coefficients remain positive. Five of the eight strength coefficients and

four of the six directional coefficients are statistically significant at the five-percent level. Thus, the empirical relationships between ex ante normalized separation and ex post model-selection evidence do not appear to be driven by the classical Vuong normalization.

G Experimental Design

G.1 Parametric Theories and Priors

We use six different parametric theories of risk preference to guide our experimental design. These functional forms, as well as the prior parameter values used to calculate the design criterion, are preregistered at <https://osf.io/xyab3/overview>. We reproduce both below. To get from the parameter priors to our design criterion, we put equal weights on the given parameter combinations.

Each model specifies binary choice probabilities between two lotteries, which we label A and B . A generic lottery L is described by a payoff vector x_L and a corresponding probability vector p_L . We use x_L^i to indicate the lottery's payout in the i 'th state, which occurs with probability p_L^i . We use $s(L)$ to denote the number of states in lottery L . For pairs of lotteries A and B in our choice problems, a common state-space is explicitly specified, i.e. $s(A) = s(B)$ and $p_A = p_B$. Additionally, all payoffs are positive in our dataset.

Expected Utility Theory (EUT)

$$\Pr(A) = \frac{1}{1 + \exp(\eta \cdot (V(B) - V(A)))},$$

where

$$V(L) = \sum_{i=1}^{s(L)} (x_L^i)^\alpha \cdot p_L^i$$

Since we never place any weight on EUT when calculating our design criterion (instead we place full weight on the nesting model CPT), we do not need to specify priors on parameters.

Cumulative Prospect Theory (CPT)

CPT builds on EUT by introducing a Prelec probability weighting function.

For a lottery L with $s(L)$ states, let $\pi_L : \{1, \dots, s(L)\} \rightarrow \{1, \dots, s(L)\}$ be the permutation that ranks outcomes in ascending order, so $x_L^{\pi_L(1)} \leq x_L^{\pi_L(2)} \leq \dots \leq x_L^{\pi_L(s(L))}$. Using the one-

parameter Prelec (1998) probability weighting function

$$w(p; \gamma) = \exp(-(-\ln p)^\gamma), \quad \gamma > 0,$$

the decision weight on the k -th ranked outcome of lottery L is the difference of weighted decumulative probabilities,

$$v_L^k(\gamma) = w\left(\sum_{j=k}^{s(L)} p_L^{\pi_L(j)}; \gamma\right) - w\left(\sum_{j=k+1}^{s(L)} p_L^{\pi_L(j)}; \gamma\right),$$

with the convention that the second sum is zero when $k = s(L)$.

As with EUT, we have

$$\Pr(A) = \frac{1}{1 + \exp(\eta \cdot (V(B) - V(A)))},$$

where now

$$V(L) = \sum_{k=1}^{s(L)} v_L^k(\gamma) \cdot (x_L^{\pi_L(k)})^\alpha.$$

Our priors cover the parameter combinations

$$\begin{aligned} (\alpha, \gamma, \eta) \in \{ & (0.6, 0.4, 0.58), \\ & (0.8, 0.4, 0.23), \\ & (0.6, 0.6, 0.73), \\ & (0.8, 0.6, 0.33), \\ & (0.6, 0.8, 0.76), \\ & (0.8, 0.8, 0.38) \} \end{aligned}$$

Random Preference Expected Utility (REU)

For each individual decision, a power utility curvature parameter α is drawn from a lognormal distribution with median m_α and log-scale σ_α ,

$$\ln \alpha \sim N(\ln m_\alpha, \sigma_\alpha^2), \quad \alpha > 0,$$

where m_α is the median curvature.²⁶

²⁶To speed computation, we truncate the distribution of α at 0.01 on the low end and 10 at the high end (with appropriate renormalization).

Given α , the lotteries are evaluated according to expected utility,

$$V(L \mid \alpha) = \sum_{j=1}^{s(L)} (x_L^j)^\alpha \cdot p_L^j,$$

and A is chosen whenever $V(A \mid \alpha) > V(B \mid \alpha)$.

To allow for dominance violations and other noise, we add a tremble: with probability $\omega \in (0, 1)$ the choice is made uniformly at random, and with probability $1 - \omega$ the subject follows their expected utility ranking. Letting $\Delta(\alpha) \equiv V(A \mid \alpha) - V(B \mid \alpha)$, the choice probability conditional on α is

$$\Pr(A \mid \alpha) = (1 - \omega) \mathbf{1}\{\Delta(\alpha) > 0\} + \frac{\omega}{2}.$$

Integrating over the distribution of α , the unconditional choice probability for a given problem is

$$\Pr(A) = (1 - \omega) \Pr_{\alpha \sim \text{LogN}(\ln m_\alpha, \sigma_\alpha^2)}[\Delta(\alpha) > 0] + \frac{\omega}{2}.$$

The estimable parameters are $(m_\alpha, \sigma_\alpha, \omega)$.

Again, we never place weight on this theory for our design criterion (all weight is on the nesting theory RCPT), so we do not specify priors.

Random Preference CPT (RCPT)

In random preference CPT (RCPT), we modify random expected utility by introducing a single (non-random) probability weighting parameter.

Given α , each lottery is evaluated according to its rank-dependent CPT value,

$$V(L \mid \alpha, \gamma) = \sum_{k=1}^{s(L)} v_L^k(\gamma) \cdot \left(x_L^{\pi_L(k)}\right)^\alpha,$$

and A is chosen whenever $V(A \mid \alpha, \gamma) > V(B \mid \alpha, \gamma)$.

As before, with probability $\omega \in (0, 1)$ the choice is made uniformly at random, and with probability $1 - \omega$ the choice matches the CPT ranking given α . Letting $\Delta(\alpha; \gamma) \equiv V(A \mid \alpha, \gamma) - V(B \mid \alpha, \gamma)$, the unconditional choice probability is

$$\Pr(A) = (1 - \omega) \Pr_{\alpha \sim \text{LogN}(\ln m_\alpha, \sigma_\alpha^2)}[\Delta(\alpha; \gamma) > 0] + \frac{\omega}{2}.$$

Our priors cover the parameter combinations

$$(m_\alpha, \sigma_\alpha, \gamma, \omega) \in \{(0.6, 0.4, 0.6, 0.4), \\ (0.8, 0.4, 0.6, 0.4), \\ (0.6, 0.6, 0.6, 0.4), \\ (0.8, 0.6, 0.6, 0.4), \\ (0.6, 0.8, 0.6, 0.4), \\ (0.8, 0.8, 0.6, 0.4)\}$$

Comparison Complexity (CC)

Under the comparison complexity model from Shubatt and Yang (2023), lotteries are evaluated according to expected utility, and binary choice noise is an increasing function of the CDF dissimilarity between the lotteries.

As in EUT, we have

$$V(L) = \sum_{i=1}^{s(L)} u(x_L^i) \cdot p_L^i,$$

where $u(x) = x^\alpha$.

Let F_L denote the CDF of lottery L over outcomes and F_L^{-1} its quantile function. The CDF-distance between lotteries A and B is

$$d^{CDF}(A, B) = \int_0^1 |u(F_A^{-1}(q)) - u(F_B^{-1}(q))| dq,$$

which measures the area between the utility-valued quantile functions of A and B . The value-dissimilarity ratio is

$$R^{CDF}(A, B) = \frac{V(A) - V(B)}{d^{CDF}(A, B)},$$

which takes values in $[-1, 1]$ by construction, with $R^{CDF} = 1$ (-1) when A (B) first-order stochastically dominates the other lottery.

Following Shubatt and Yang (2026), the choice probability is a monotone transformation of R^{CDF} ,

$$\Pr(A) = G(R^{CDF}(A, B; \alpha)),$$

with

$$G(r) = \begin{cases} (1 - \kappa) - (0.5 - \kappa)(1 - r)^\gamma & \text{if } r \geq 0, \\ \kappa + (0.5 - \kappa)(1 + r)^\gamma & \text{if } r < 0, \end{cases}$$

where $\kappa \in [0, 1/2)$ is a tremble parameter governing error rates at dominance and $\gamma > 0$ governs the curvature of the response function in the ratio.

The estimable parameters are (α, γ, κ) .

Our priors cover the parameter combinations

$$\begin{aligned} (\alpha, \kappa, \gamma) \in \{ & (0.6, 0.03, 0.5), \\ & (0.8, 0.03, 0.5), \\ & (0.6, 0.1, 0.5), \\ & (0.8, 0.1, 0.5), \\ & (0.6, 0.03, 0.85), \\ & (0.8, 0.03, 0.85), \\ & (0.6, 0.1, 0.85), \\ & (0.8, 0.1, 0.85) \} \end{aligned}$$

Salience (SAL)

We use the continuous version of the salience model of choice under risk from Bordalo et al. (2013). When a lottery is evaluated according to salience, the decision weight placed on each state is distorted according to how *salient* that state is.

The salience value for a lottery A in the menu $\{A, B\}$ is

$$V(A; B) = \sum_{i=1}^{s(A)} \omega_i(A, B) \cdot p_i \cdot x_A^i,$$

where $\omega_i(A, B)$ is a salience distortion term given by

$$\omega_i(A, B) = \frac{\delta^{-\sigma(x_A^i, x_B^i)}}{\sum_{j=1}^{s(A)} \delta^{-\sigma(x_A^j, x_B^j)} p_j},$$

and σ is the symmetric salience function²⁷

$$\sigma(y, z) = \frac{|y - z|}{|y| + |z| + 1}$$

Assuming logit noise,

$$\Pr(A) = \frac{1}{1 + \exp(\eta \cdot (V(B; A) - V(A; B)))}.$$

Our priors cover the parameter combinations

$$(\delta, \eta) \in \{(0.01, 0.042), \\ (0.1, 0.086)\}$$

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G.2 Experimental Instructions

²⁷In Bordalo et al. (2012), the salience function takes a parameter θ ; we have restricted this value to 1, due to weak identification. In pilot data, allowing θ to be estimated offered only very minor model fit improvements.

²⁸We roughly calibrated our prior on δ using the common consequence problem studied in Section V of Bordalo et al. (2012).

Instructions I: Overview

Please read these instructions carefully. There will be comprehension checks. If you fail these checks, you will be excluded from the study and you will not receive the completion payment.

In this study, you will make multiple decisions.

Your payment will consist of two components:

- Completion fee:
If you pass all our **comprehension checks and complete the study**, you will receive a **completion fee of \$4.50**.
- Additional bonus:
In each of 50 rounds, you will make a decision. One of the decisions will be selected at random by the computer, with equal probability, and will determine your **bonus**. The **maximal bonus** you can earn in this study is **\$50**. Your bonus will actually be paid out to you with probability 0.1.

Instructions II: Lottery Pairs

Throughout this study, you will encounter pairs of lotteries. The lotteries, called **Option A** and **Option B**, will each pay out money depending on the color of a randomly drawn ball. We will present these lotteries to you in a table like the one below:

Probability		Option A	Option B
	40%	\$12	\$24
	50%	\$60	\$89
	10%	\$68	\$11

Instructions II: Lottery Pairs

The first two columns tell us the *likelihood* of drawing each colored ball. For each lottery pair, there are 20 balls total, one of which will be drawn at random to determine payouts.

Probability		Option A	Option B
	40%	\$12	\$24
	50%	\$60	\$89
	10%	\$68	\$11

Instructions II: Lottery Pairs

The first column depicts the number of red, green, and blue balls.

	Probability		Option A	Option B
This problem has 8 red balls...		40%	\$12	\$24
...10 green balls...		50%	\$60	\$89
...and 2 blue		10%	\$68	\$11

Instructions II: Lottery Pairs

The second column expresses the same information as probabilities.

	Probability		Option A	Option B
Red will be drawn 40% of the time		40%	\$12	\$24
Green will be drawn 50% of the time		50%	\$60	\$89
Blue will be drawn 10% of the time		10%	\$68	\$11

Instructions II: Lottery Pairs

The third and fourth columns tell us how much the options pay out for each ball color.

Probability		Option A	Option B	
	40%	\$12	\$24	If red is drawn, A pays \$12, B pays \$24
	50%	\$60	\$89	If green is drawn, A pays \$60, B pays \$89
	10%	\$68	\$11	If blue is drawn, A pays \$68, B pays \$11

Instructions III: Study Tasks

Task 1

For each pair of lotteries you encounter, you will be asked to **click on the option you would prefer to receive**. For example, if you liked **Option B** better than **Option A**, you would click the button for **Option B**.

Probability		Option A	Option B
	40%	\$12	\$24
	50%	\$60	\$89
	10%	\$68	\$11

Which option do you choose?
Please select one.

There are no right or wrong answers. We are just interested to know which option you would prefer.

Instructions IV: Attention Checks

Throughout the study, you will encounter attention checks. Be sure to read each page carefully in order to pass these checks. If you fail an attention check, you will not be eligible for a bonus payment.

Towards the end of the survey, you will be specifically asked to enter the **key phrase** from the instructions. The key phrase is **"roses are red"**. We recommend noting this down so that you can access it later.

Instructions V: Bonus Payment

If you complete the entire study and pass all attention checks, you will have the chance to earn a bonus.

If you are selected for a bonus, the computer will randomly select **one of the 50 choices** you saw and simulate drawing a red, green, or blue ball, according to the stated probabilities. If a red ball is drawn, you will actually receive the "red" payment for the option you chose; if a green ball is drawn, you will receive the "green" payment, and so on.

On average, **these bonus payments are worth \$25**. If you successfully complete the study, you'll have a **1 in 10 chance** of being selected for a bonus.

H Appendix: Heterogeneity

Beyond selecting a single best-fitting model class, researchers also use lab experiments to study population-level heterogeneity: estimating the share of subjects whose behavior is consistent with each candidate model class. We adapt the design problem to this goal.

When researchers propose a set of candidate models and examine, across a sample of subjects, the share whose behavior is best described by each, they implicitly posit a mixture data-generating process at the population level. We make this assumption explicit by considering distributions in $\mathcal{P}$ of the hierarchical-mixture (random-coefficients) form: each subject i has a latent type $Z_i \sim \text{Mult}(w^0)$ with shares $w^0 \in \Delta^{K-1}$; conditional on $Z_i = k$, an individual parameter $\theta_i \sim F_k(\cdot \mid \eta_k^0)$ drawn from a parametric within-type distribution F_k with hyperparameter $\eta_k^0 \in H_k$ (the functional form of F_k is taken as known, e.g., $F_k = \mathcal{N}(\mu_k, \Sigma_k)$ on transformed parameters with $\eta_k = (\mu_k, \Sigma_k)$); conditional on (Z_i, θ_i) , $Y_{it} \sim g_k(\cdot \mid X_t, \theta_i)$ independently across t . The implied per-task density is

$$p_0(\cdot \mid X_t) = \sum_{k=1}^K w_k^0 \int_{\Theta_k} g_k(\cdot \mid X_t, \theta) F_k(d\theta \mid \eta_k^0), \quad (w^0, \eta_1^0, \dots, \eta_K^0) \in \Delta^{K-1} \times \prod_k H_k.$$

Assumption 1 (Heterogeneity prior and loss).

- (i) *Prior. The hierarchical mixture prior Π on $\mathcal{P}$ is induced by $w^0 \sim \text{Dir}(\alpha)$ on Δ^{K-1} and, independently, $\eta_k^0 \sim \pi_{\eta_k}$ on H_k for each $k \in \mathcal{K}$.*
- (ii) *Loss. The action space is $\mathcal{A} = \Delta^{K-1}$: the experimenter reports a mixture-weight vector $\hat{w}$, with squared-error loss*

$$\ell(a, p_0) = \|a - w(p_0)\|^2,$$

where $w(p_0) \in \Delta^{K-1}$ is the mixture-weight vector underlying p_0 .

Although the Bayes-optimal action under Assumption 1 is the posterior mean of w^0 , this is rarely the procedure used in practice. We instead analyze two inference exercises common in the empirical literature: *individual classification* and *mixture estimation* (Hey, 2001; Bruhin et al., 2010; Harrison et al., 2010; Königsheim et al., 2019; Cavagnaro et al., 2013; Bayrak and Hey, 2020; McGranaghan et al., 2024). In individual classification, the researcher first assigns each subject to the best-fitting or least-violated model class using only that subject’s T -task data, and then reports the share of subjects assigned to each class. In mixture estimation, the researcher pools data across subjects and jointly estimates the population shares w and the within-class parameters by maximizing the mixture likelihood.

Individual classification. We formalize the individual classification procedure as follows.²⁹ For each subject i and each candidate class $k \in \mathcal{K}$, compute the within-class MLE on that subject's own T tasks,

$$\hat{\theta}_{k,i} = \arg \max_{\theta \in \Theta_k} \sum_{t=1}^T \log g_k(Y_{it} | X_t, \theta),$$

i.e., the parameter value that best fits subject i 's data assuming subject i is of type k . Next, classify subject i as the type whose best-fitting parameter rationalizes the data best,

$$\hat{Z}_i = \arg \max_{k \in \mathcal{K}} \sum_{t=1}^T \log g_k(Y_{it} | X_t, \hat{\theta}_{k,i}),$$

where $\hat{Z}_i \in \mathcal{K}$ is the assigned type for subject i . Finally, aggregate the assignments into the empirical type shares,

$$\hat{w}_k^{\text{cls}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\hat{Z}_i = k\},$$

the fraction of subjects classified as type k . The evaluator's decision rule is given by $\hat{w}^{\text{cls}} = (\hat{w}_1^{\text{cls}}, \dots, \hat{w}_K^{\text{cls}})$. The following proposition shows that the power of design governs the statistical properties of the $\hat{w}^{\text{cls}}$ estimator.

Proposition 5 (Bias in individual classification).

Let Q be the $K \times K$ confusion matrix with $Q_{kk'} = \Pr(\hat{Z}_i = k' | Z_i = k)$, averaged over $\theta_i \sim F_k$. As $N \rightarrow \infty$, $\hat{w}^{\text{cls}} \xrightarrow{P} Q^\top w^0$, and as $T \rightarrow \infty$ each entry of Q is a $(K-1)$ -dimensional Gaussian orthant probability,

$$Q_{kk'} = \mathbb{E}_{\theta \sim F_k} [\Phi(\sqrt{T} \tilde{\rho}^{(k,k')}(\theta))], \quad \tilde{\rho}_j^{(k,k')}(\theta) = \frac{\bar{D}_{k,j}(\theta) - \bar{D}_{k,k'}(\theta)}{\sqrt{\tilde{V}_j^{(k,k')}(\theta)}}, \quad j \neq k', \quad (28)$$

where $\bar{D}_{k,j}(\theta)$ is from Definition 2 (with $\bar{D}_{k,k} \equiv 0$), $\tilde{V}_j^{(k,k')}(\theta)$ is the design-averaged variance of $\log[g_{k'}(\cdot | X_t, \theta_{k'}^*(\theta)) / g_j(\cdot | X_t, \theta_j^*(\theta))]$ under $Y \sim g_k$, and Φ is taken under the induced

²⁹In practice, researchers do not always use maximum likelihood in the first-step classification — some rely on heuristic classification rules instead. We abstract from these differences; Proposition 5 below holds whenever the first-step classifier is imperfect.

correlation matrix; the diagonal recovers $\tilde{\rho}^{(k,k)} = \bar{\rho}_k$. Consequently,

$$\text{MSE}(\hat{w}_k^{\text{cls}}) = \underbrace{\frac{r_k(1-r_k)}{N}}_{\text{variance: } O(1/N)} + \underbrace{\left((Q^\top w^0)_k - w_k^0\right)^2}_{\text{bias}^2: \text{persists in } N}, \quad r_k = (Q^\top w^0)_k, \quad (29)$$

and the design problem (7) reduces in the large- N limit to $X^* \in \arg \min_{X \in \mathcal{X}_\ell^T} \mathbb{E}[\|(Q^\top - I)w^0\|^2]$ under the prior $\pi_w \otimes \bigotimes_k \pi_k$.

Proof. See Appendix I.1. □

The same effect size $\bar{\rho}_k(\theta)$ that governed the probability of correct selection in Proposition 1 now governs the entries of the confusion matrix. When design power is high, Q is close to the identity: individual classification is accurate, and the estimated population shares are approximately unbiased. When design power is low, $Q_{kk} \rightarrow 1/K$ and the estimated collapse toward the uniform vector $\frac{1}{K}\mathbf{1}$. This collapse occurs regardless of the truth w_0 , and cannot be mitigated by larger sample size. The variance of the share estimates vanishes as N increases, but the bias towards $\frac{1}{K}$ persists if the design itself is not powerful enough to classify subjects accurately.

I Appendix: Proofs

I.1 Proof of Proposition 1 and 5

Both proofs maintain the following regularity conditions: each Θ_k is compact with non-empty interior; $\log g_k(y | x, \theta)$ is twice continuously differentiable in θ with envelopes integrable under each candidate true law; the pseudo-true parameter $\theta_j^*(\theta_k, X)$ is uniquely identified on every admissible design; and the LLR variances $\bar{V}_{k,j}(\theta_k)$ are finite with $\bar{D}_{k,j}(\theta_k) > 0$ for every $j \neq k$ (separation). For Proposition 5 we additionally impose the no-dominant-task Lindeberg condition: for each (k, j) and $\theta \in \Theta_k$,

$$\frac{\max_{t \leq T} V_{t,kj}(\theta)}{\sum_{t \leq T} V_{t,kj}(\theta)} \rightarrow 0 \quad \text{as } T \rightarrow \infty. \quad (30)$$

Proof of Proposition 1. Condition on the realized truth $(k^0, \theta_{k^0}^0) \sim \Pi$ and treat T as fixed. For each subject i , define the $(K-1)$ -vector

$$\Lambda_i = \left(\frac{1}{T} \sum_{t=1}^T \log \frac{g_{k^0}(Y_{it} | X_t, \theta_{k^0}^0)}{g_j(Y_{it} | X_t, \theta_j^*(\theta_{k^0}^0, X))} \right)_{j \neq k^0}.$$

Subjects are i.i.d., so $(\Lambda_i)_{i \leq N}$ is i.i.d. with $\mathbb{E}[\Lambda_i] = \bar{D}_{k^0}(\theta_{k^0}^0)$ and, by conditional independence of Y_{it} across t , $\text{Cov}(\Lambda_i) = \frac{1}{T} \bar{\Sigma}_{k^0}$ where $(\bar{\Sigma}_{k^0})_{jj'} = \bar{C}_{k^0, jj'}$. By the regularity conditions, the within-class MLE $\hat{\theta}_k \xrightarrow{p} \theta_k^*$ and the plug-in error in the sample log-likelihood is $O_p(1) = o_p(\sqrt{NT})$. Any selection rule with $o(\sqrt{NT})$ penalty therefore satisfies $\{\hat{k} = k^0\} = \bigcap_{j \neq k^0} \{\bar{\Lambda}_j > 0\} + o_p(1)$, where $\bar{\Lambda} = N^{-1} \sum_i \Lambda_i$. The multivariate CLT in N (with T fixed and finite second moments) gives

$$\sqrt{NT}(\bar{\Lambda} - \bar{D}_{k^0}) \xrightarrow{d} \mathcal{N}_{K-1}(0, \bar{\Sigma}_{k^0}).$$

Standardizing by $(\sqrt{\bar{V}_{k^0, j}})$ converts $\bar{\Sigma}_{k^0}$ to the correlation matrix $\bar{R}_{k^0}$ and the mean to $\bar{\rho}_{k^0}$. By symmetry of the multivariate normal, $\Pr(\hat{k} = k^0) \rightarrow \Pr(W \leq \sqrt{NT} \bar{\rho}_{k^0})$ for $W \sim \mathcal{N}(0, \bar{R}_{k^0})$, which is $\Phi(\sqrt{NT} \bar{\rho}_{k^0}(\theta_{k^0}^0))$. Averaging over (P_k, π_k) gives the displayed objective. $\square$

Proof of Proposition 5. Condition on subject i with $(Z_i = k, \theta_i = \theta)$. The within-subject MLE applied to subject i 's T tasks satisfies $\hat{\theta}_{m,i} \xrightarrow{p} \theta_m^*(\theta, X)$ as $T \rightarrow \infty$ for each m , with plug-in error $O_p(1) = o_p(\sqrt{T})$. Hence $\{\hat{Z}_i = k'\}$ is equivalent (up to $o_p(\sqrt{T})$) to the joint inequality

$$\frac{1}{T} \sum_{t=1}^T \log \frac{g_{k'}(Y_{it} | X_t, \theta_{k'}^*(\theta))}{g_j(Y_{it} | X_t, \theta_j^*(\theta))} > 0 \quad \text{for all } j \neq k'.$$

Under $Y_{it} \sim g_k(\cdot | X_t, \theta)$, the j -th LLR has design-averaged mean $\bar{D}_{k,j}(\theta) - \bar{D}_{k,k'}(\theta)$ (with $\bar{D}_{k,k} \equiv 0$) and design-averaged variance $\tilde{V}_j^{(k,k')}(\theta)$. The Lindeberg condition (30) licenses the multivariate Lindeberg-Feller CLT in T , so the standardized $(K-1)$ -vector converges to $\mathcal{N}_{K-1}(0, \tilde{R}^{(k,k')}(\theta))$, giving

$$\Pr(\hat{Z}_i = k' | Z_i = k, \theta_i = \theta) \rightarrow \Phi(\sqrt{T} \tilde{\rho}^{(k,k')}(\theta)).$$

Integrating against F_k yields (28); the diagonal $k' = k$ has $\bar{D}_{k,j} - \bar{D}_{k,k} = \bar{D}_{k,j}$, so $\tilde{\rho}^{(k,k)} = \bar{\rho}_k$.

Across subjects, $\mathbf{1}\{\hat{Z}_i = k\}$ are i.i.d. Bernoulli(r_k) with $r_k = \sum_j w_j^0 Q_{jk} = (Q^\top w^0)_k$. The LLN gives $\hat{w}_k^{\text{cls}} \xrightarrow{p} r_k$, with bias $r_k - w_k^0 = ((Q^\top - I)w^0)_k$ and variance $r_k(1 - r_k)/N$, yielding (29). Substituting into (7) with the squared loss $\|a - w(p_0)\|^2$ gives $X^* \in \arg \min_X \mathbb{E}[\|(Q^\top - I)w^0\|^2]$. $\square$

J Implementation details

J.1 Implementation of computer experiments

J.2 Discussion of possible degeneracy of objective

When researchers optimize for experiments with complicated model classes, it may not be clear to the researcher how one should specify parameters that lead to non-overlapping models. There are well-known complications around the limiting distribution of the likelihood ratio test for generic overlapping models (?). Simply put, when the conditioned true model is within the intersection of two model classes, the variance in Proposition 1 becomes degenerate. Therefore, in our implementation, we apply a modified likelihood-ratio test following the spirit of Schennach and Wilhelm (2017).

Suppose we have K models with likelihoods $g_k(Y, \theta_k)$. Proposition 1 and its proof in Appendix X relies on the model selection statistic

$$T = \left(\frac{1}{n} \sum_i \log \frac{g_1(Y_i, \hat{\theta}_1)}{g_2(Y_i, \hat{\theta}_2)}, \dots, \frac{1}{n} \sum_i \log \frac{g_1(Y_i, \hat{\theta}_1)}{g_K(Y_i, \hat{\theta}_K)} \right)$$

and its limiting distribution conditional on $g_1(\theta_1)$ being the true density. Note that when $g_1(\theta_1) = g_2(\theta_2^*)$, the first coordinate is almost surely 0, and hence the covariance matrix is degenerate. To avoid this issue, we implement the following model selection rule.

Consider a sequence of weights $w_i^{(k)}$ for $i = 1, \dots, n; k = 1, \dots, K$. As before, we let $\tilde{L}_k = \frac{1}{n} \sum_i w_i^{(k)} \log g_k(Y_i, \hat{\theta}_k)$ where $\hat{\theta}_k$ is obtained by the standard maximum likelihood, and $\hat{\theta}_k \rightarrow_p \theta_k^*$ converges to the pseudo-true parameter relative to p_0 . The model selection rule chooses model 1 iff $\tilde{L}_1 > \tilde{L}_k$ for $k \neq 1$. This is very similar to what we have before, except that each observation is now weighted by $w_i^{(k)}$. One could see that if the weights are equal to 1, then we have exactly what we had before. The aim is to choose the weights cleverly to avoid degeneracy in the analogous T statistic.

Assume n is a multiple of K . Denote $\ell(Y_i) = (\log g_1(Y_i, \hat{\theta}_1), \dots, \log g_K(Y_i, \hat{\theta}_K))'$. Let $D_i = \text{diag}(w_i^{(1)}, \dots, w_i^{(K)})$, and $\tilde{L} = \frac{1}{n} \sum_i D_i \ell(Y_i) = (\tilde{L}_1, \dots, \tilde{L}_K)'$. Note that $\tilde{T} = (\tilde{L}_1 - \tilde{L}_2, \dots, \tilde{L}_1 - \tilde{L}_K)' = C\tilde{L}$ where

$$C = \begin{bmatrix} 1 & -1 & 0 & \dots & 0 \\ 1 & 0 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \dots & -1 \end{bmatrix}_{(K-1) \times K}$$

We will choose $w_i^{(k)}$ such that $w_i^{(k)} = 1 + \varepsilon u_{r(i)+k}$, where $r(i) \equiv (i-1) \pmod K$, and the index $r(i) + k$ is interpreted as $z \pmod K$, i.e., in $\{1, \dots, K\}$. The sequence $u = (u_1, \dots, u_K)$ satisfies $\sum_{m=1}^K u_m = 0$. Example: for $K = 3$, let $u = (0, -1, 1)$

i	1	2	3	4	5	6
$r(i)$	0	1	2	0	1	2
$w_i^{(1)}$	1	$1 + \varepsilon$	$1 - \varepsilon$	1	$1 + \varepsilon$	$1 - \varepsilon$
$w_i^{(2)}$	$1 - \varepsilon$	1	$1 + \varepsilon$	$1 - \varepsilon$	1	$1 + \varepsilon$
$w_i^{(3)}$	$1 + \varepsilon$	$1 - \varepsilon$	1	$1 + \varepsilon$	$1 - \varepsilon$	1

First observe that because $\hat{\theta}_k \rightarrow_p \theta_k^*$, we can characterize $E[\ell(Y_i)] \rightarrow \mu$ and $\text{Cov}[\ell(Y_i)] \rightarrow \Sigma$ under large n .

Next we will show that by CLT and our construction, $\sqrt{n}(\tilde{L} - \mu) \rightarrow N(0, \Sigma + \varepsilon^2(\Sigma \circ R))$, R is a $K \times K$ matrix and $R_{ab} = S_{b-a}$ where $S_h = \frac{1}{K} \sum_{m=1}^K u_m u_{m+h}$ is the circular autocorrelation, again indices interpreted in terms of $\pmod K$.

To see the expectation statement, observe that $E[\tilde{L}] = \frac{1}{n} \sum_i D_i E[\ell(Y_i)]$. D_i is a function of i only through $r(i)$, and since n is a multiple of K , we have

$$E[\tilde{L}] = \frac{1}{n} \sum_i D_i E[\ell(Y_i)] = \frac{1}{n} \frac{n}{K} \sum_{r=0}^{K-1} D_r E[\ell(Y_i)] = E[\ell(Y_i)] \rightarrow \mu.$$

To see the variance statement, observe that

$$\text{Var}(\sqrt{n}\tilde{L}) = \frac{1}{n} \sum_i D_i \text{Var}(\ell(Y_i)) D_i = \frac{1}{K} \sum_{r=0}^{K-1} D_r \Sigma D_r$$

Element-wise, we have

$$[D_r \Sigma D_r]_{ab} = \sigma_{ab} (1 + \varepsilon u_{r+a} + \varepsilon u_{r+b} + \varepsilon^2 u_{r+a} u_{r+b}).$$

As we sum over r , we have $\sum_{r=0}^{K-1} u_{r+a} = 0$ for any $a = 1, \dots, K$, by construction, so the linear in ε terms cancel out. Hence

$$\left[\frac{1}{K} \sum_{r=0}^{K-1} D_r \Sigma D_r \right]_{ab} = \sigma_{ab} + \varepsilon^2 \frac{1}{K} \sum_{r=0}^{K-1} u_{r+a} u_{r+a+(b-a)} = \sigma_{ab} + \varepsilon^2 R_{ab}.$$

So we have $\tilde{\Sigma} = \Sigma + \varepsilon^2(\Sigma \circ R)$.

Our implemented model selection is to choose model 1 iff $\tilde{T} = C\tilde{L} > 0$ on all dimensions.

This is

$$Pr(C\tilde{L} > 0) = Pr(C\tilde{\Sigma}^{1/2}Z > -\sqrt{n}C\mu).$$